



电路基础 期中复习

Chapter 1-8

针对课程：EE104/SME103 电路基础

自我介绍

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微电子科学与工程系

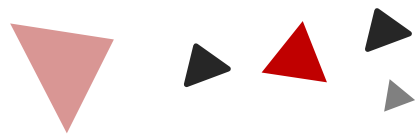
欢迎同学们加入深港微电子学院！

Content 期中之前所学内容

Chapter1. Basic Concept about circuits	电路基本概念
Chapter2. Basic Laws	电路基本定则
Chapter3. Methods of Analysis	电路分析方法
Chapter4. Circuit Theorems	电路定理
Chapter5. Operational Amplifiers	运算放大器
Chapter6. Capacitors and Inductors	电容和电感
Chapter7. First-order Circuit	一阶微分电路
Chapter8. Second-order Circuit	二阶微分电路

电路基本概念

1



Chapter

1.1 电荷，电流，电压和功率

1.2 Associated and unassociated reference directions

无源和有源参考方向

1.3 国际单位制

1.4 Independent sources and Dependent sources

独立源/受控源

1.1&1.2 电荷，电流，电压和功率 & 无源（有源）符号约定

$$i \triangleq \frac{dq}{dt}$$

$$Q \triangleq \int_{t_0}^t i dt$$

$$v_{ab} \triangleq \frac{dw}{dq}$$

$$p \triangleq \frac{dw}{dt}$$

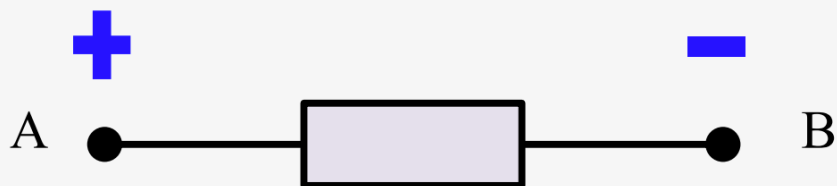
$$p = vi$$

$P = ui > 0$ absorb energy
 $P = ui < 0$ supply energy

$P = ui > 0$ supply energy
 $P = ui < 0$ absorb energy

Associated reference directions

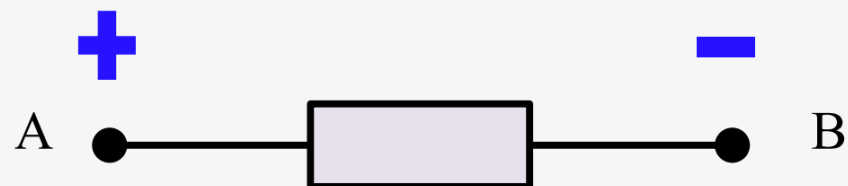
Reference voltage direction



➔ Reference current direction

Nonassociated reference directions

Reference voltage direction



➔ Reference current direction

1.3 国际单位制

需要记住的：

G (10^9)

M (10^6)

K (10^3)

m (10^{-3})

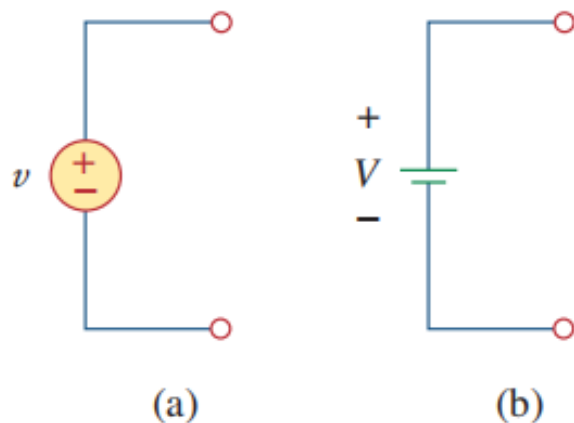
μ (10^{-6})

n (10^{-9})

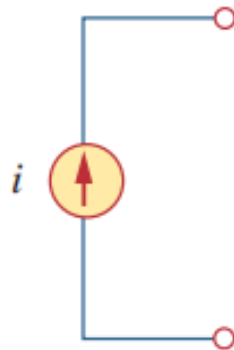
p (10^{-12})

Prefix	Symbol for Prefix		Scientific Notation
exa	E	1 000 000 000 000 000 000	10^{18}
peta	P	1 000 000 000 000 000	10^{15}
tera	T	1 000 000 000 000	10^{12}
giga	G	1 000 000 000	10^9
mega	M	1 000 000	10^6
kilo	k	1 000	10^3
hecto	h	100	10^2
deka	da	10	10^1
----	--	1	10^0
deci	d	0.1	10^{-1}
centi	c	0.01	10^{-2}
milli	m	0.001	10^{-3}
micro	μ	0.000 001	10^{-6}
nano	n	0.000 000 001	10^{-9}
pico	p	0.000 000 000 001	10^{-12}
femto	f	0.000 000 000 000 001	10^{-15}
atto	a	0.000 000 000 000 000 001	10^{-18}

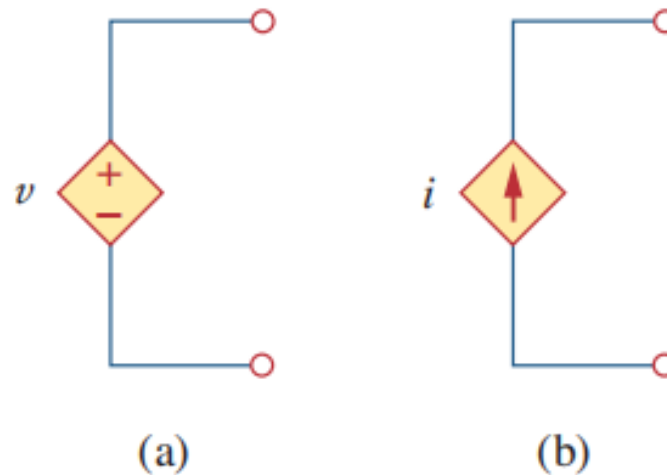
1.4 独立源和受控源



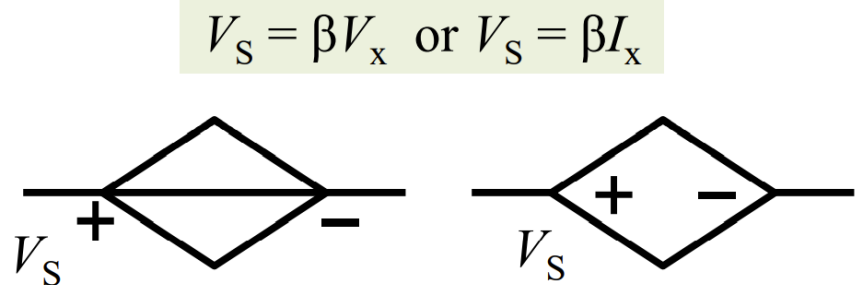
独立电压源



独立电流源

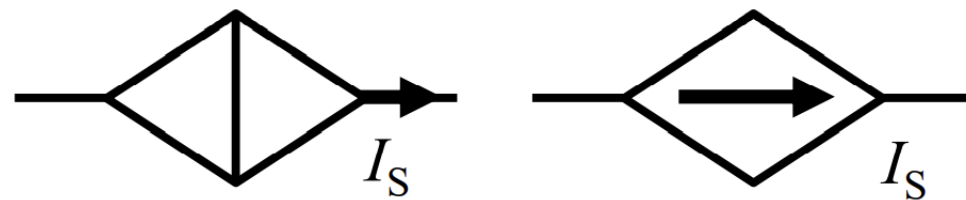


受控源



受控电压源

$$I_S = \beta V_x \text{ or } I_S = \beta I_x$$



受控电流源

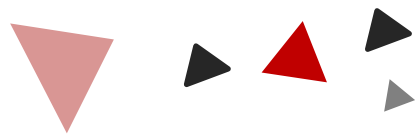
1.4 独立源和受控源的性质

性质1：独立源的状态与电路无关，即无论电路出于什么状态，其提供的电压（电流）都为恒定值；而受控源的所提供的电压（电流）取决于电路。

性质2：理想独立电压源的内阻为零；理想电流源内阻为无穷大；而受控源的内阻会随电路改变。

电路基本定律

2



Chapter

2.1 支路，节点，回路和网孔

2.2 KCL and KVL

2.3 等效电阻

2.4 Δ -Y and Y- Δ

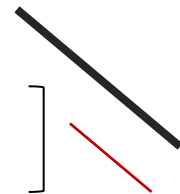
2.1 支路，节点，回路和网孔

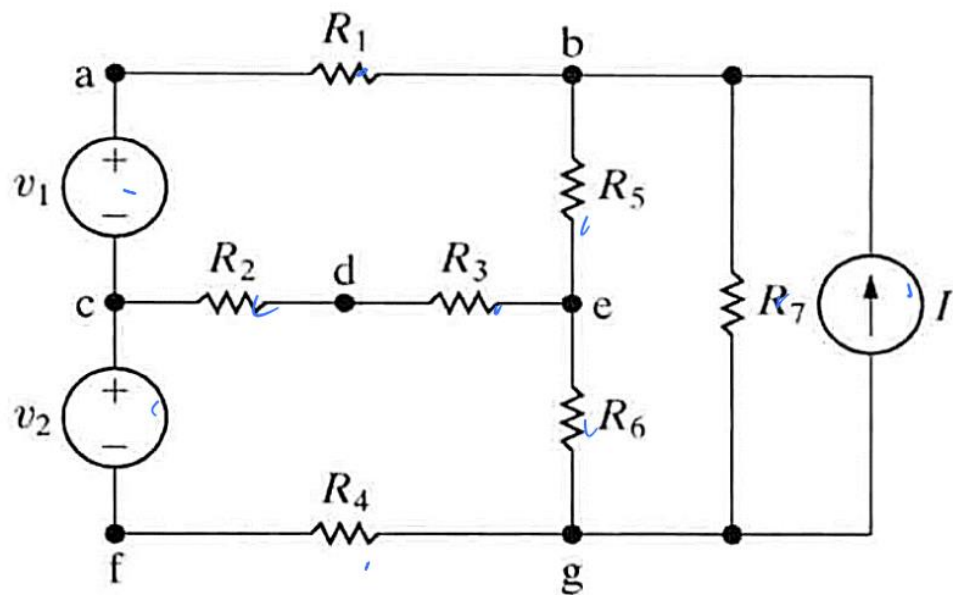
支路： 器件个数

节点： 两条或多条支路的连接点，一根导线（导线上没有元件）
连接的两个节点合并为一个节点。

网孔： 不含有其它闭合电路的回路。

回路： 任意闭合路径。





$$b = m + n - 1$$

Solution:

(a)	Nodes	$n = 7$	a, b, c, d, e, f, g
(b)	Branches	$b = 10$	$v_1, v_2, R_1, R_2, R_3, R_4, R_5, R_6, R_7, I$
(c)	Meshes	$m = 4$	$(1) v_1 - R_1 - R_5 - R_3 - R_2$ $(2) v_2 - R_2 - R_3 - R_6 - R_4$ $(3) R_5 - R_7 - R_6$ $(4) R_7 - I$
(d)	2 loops (not meshes)	--	$(1) v_1 - R_1 - R_5 - R_6 - R_4 - v_2$ $(2) I - R_5 - R_6$

2.2 基尔霍夫电压（电流）定律

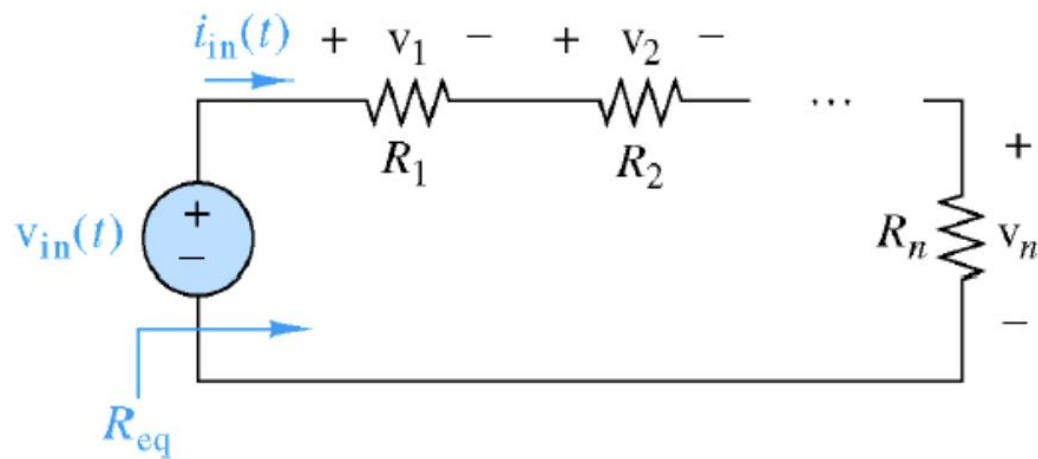
$$\sum i_{\text{in}} = \sum i_{\text{out}}$$

$$\sum_{m=1}^M v_m = 0$$

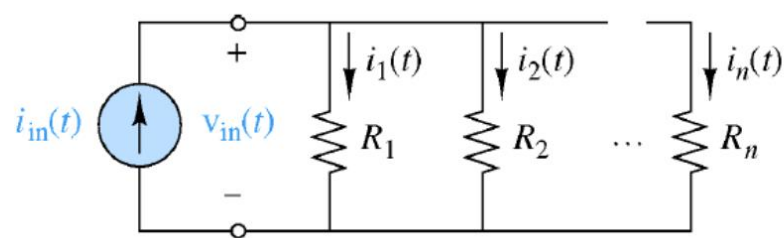
电流进入节点为正，离开为负。

从正极到负极通过电压表，电压下降，反之上升。

2.3 分压和分流



$$v_j(t) = \frac{R_j}{R_1 + R_2 + \dots + R_n} v_{in}(t)$$



$$i_j(t) = \frac{G_j}{G_1 + G_2 + \dots + G_n} i_{in}(t)$$

$$i_1 = \frac{R_2 i}{R_1 + R_2}, \quad i_2 = \frac{R_1 i}{R_1 + R_2}$$

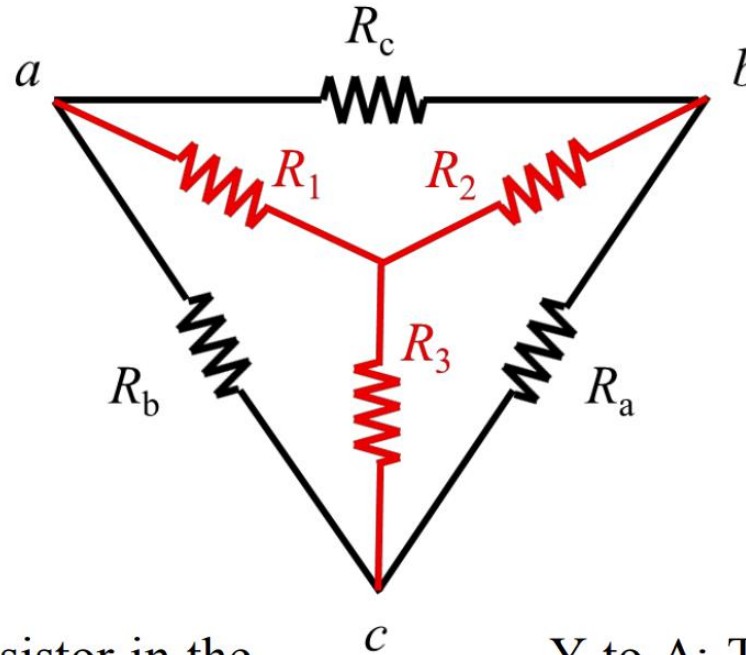
2.4 Δ -Y 转换

Δ to Y

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$$

$$R_2 = \frac{R_a R_c}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_b R_a}{R_a + R_b + R_c}$$



Y to Δ

$$R_a = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_1}$$

$$R_b = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_2}$$

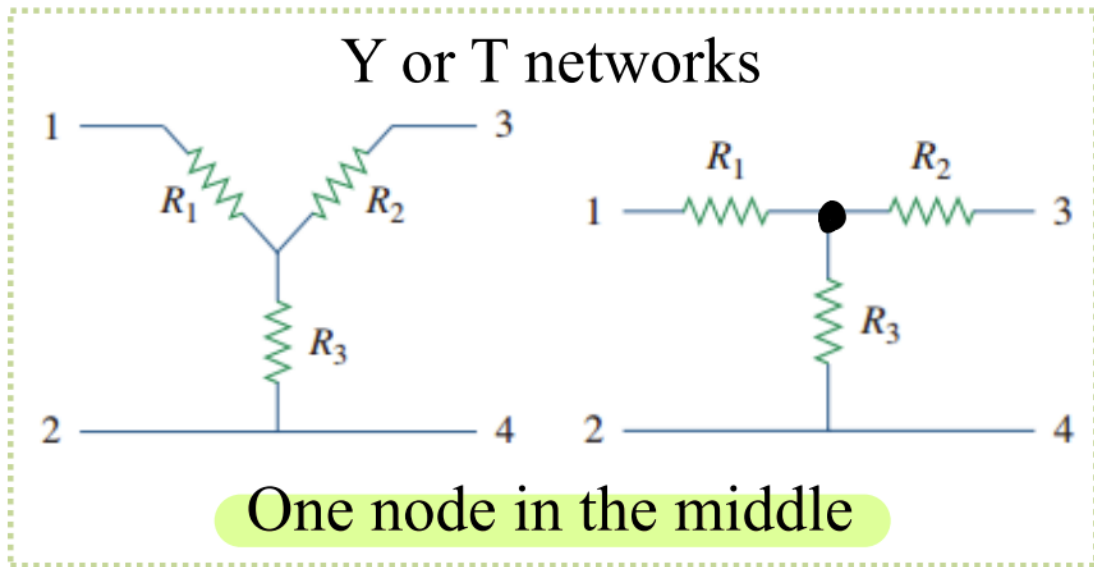
$$R_c = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_3}$$

Δ to Y: the resistance of a resistor in the Y network is the product of the resistance of the two adjacent resistors in the Δ network, divided by the sum of the resistance of the three Δ resistors.

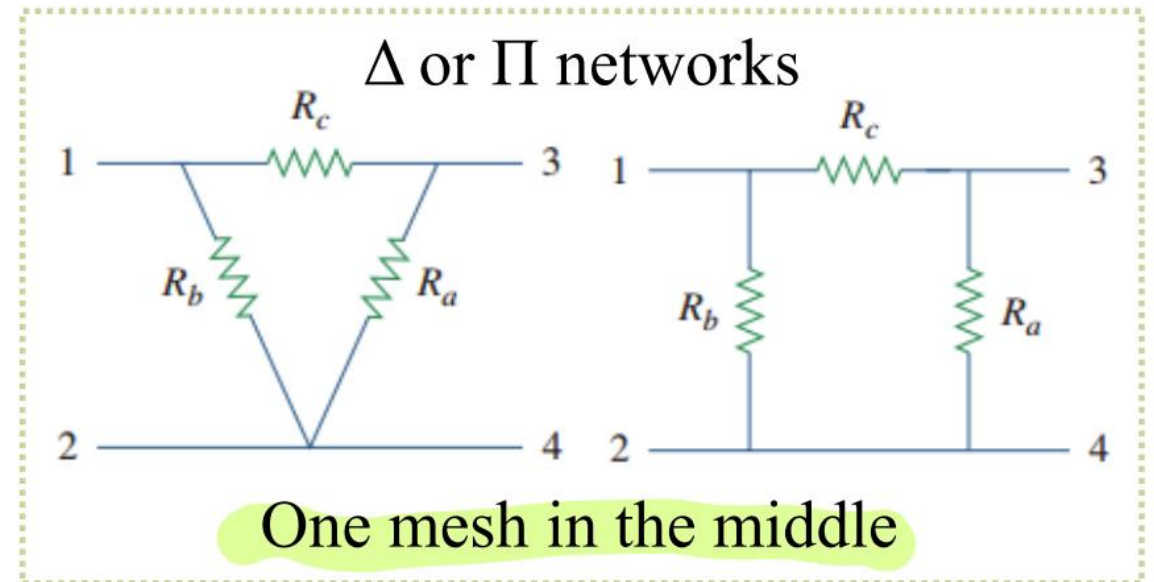
Y to Δ : The resistance of a resistor in the Δ network is the sum of all possible products the resistance of the Y resistors taken two at a time, divided by the resistance of the opposite Y resistor.

2.4 Δ -Y 转换

常见Y电路：

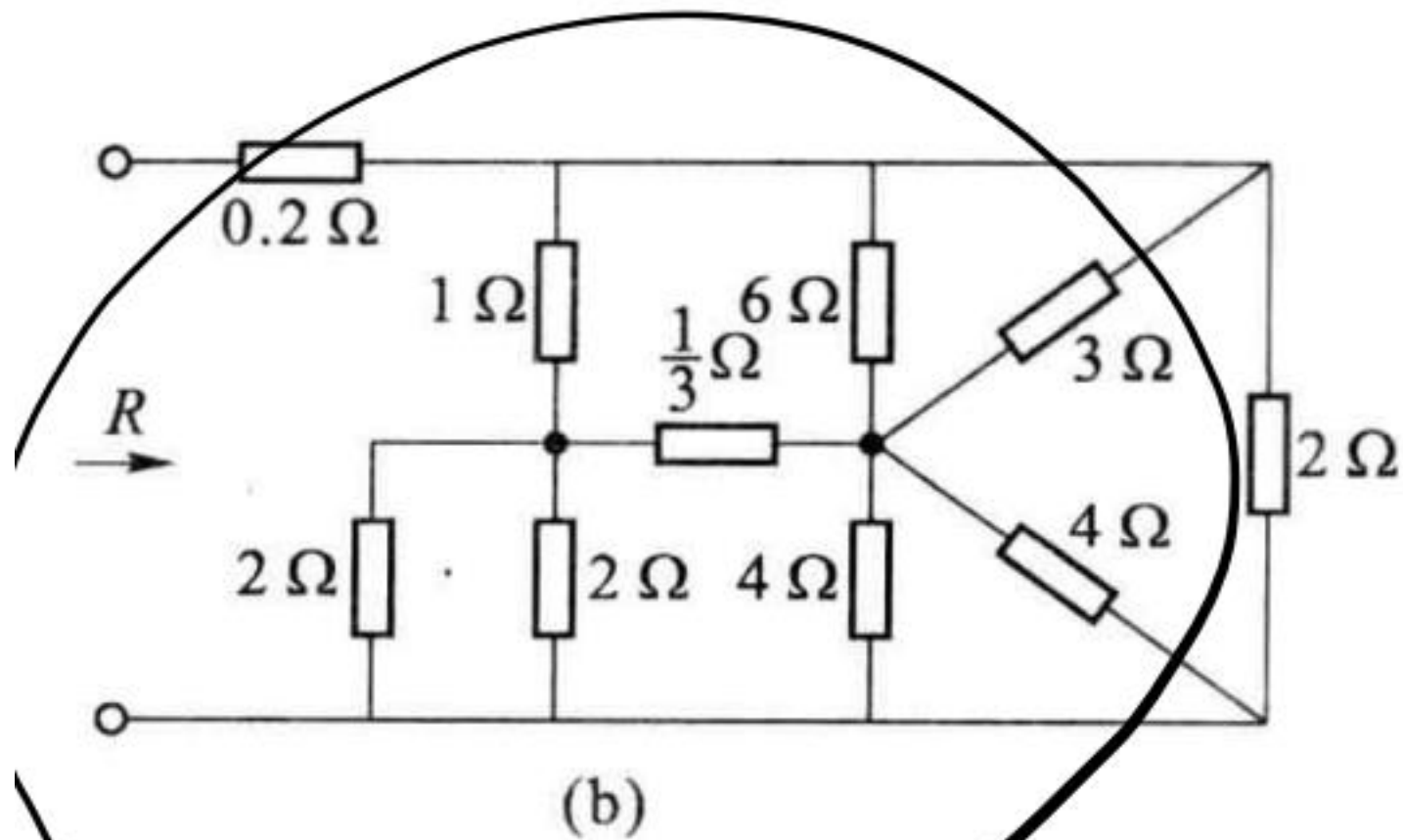


常见 Δ 电路：



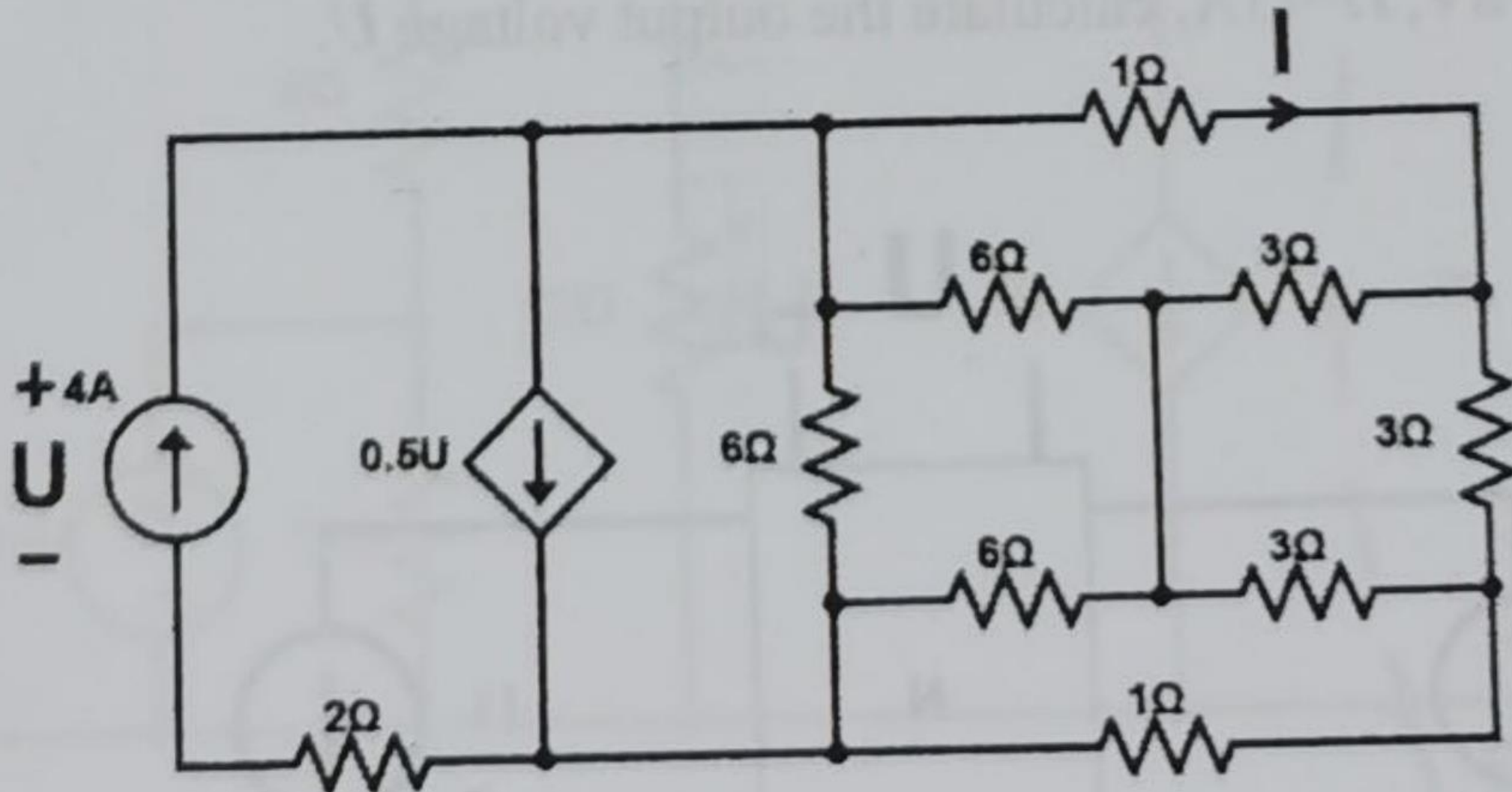
转换的目的：化简电路，减少支路

(其他教材) 求解等效电阻



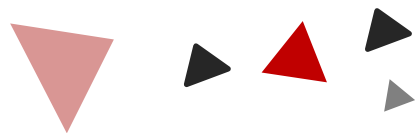
图题 2.9

(2021期中) 求解 I



电路的系统分析方法

3



Chapter

3.1 Nodal voltage analysis

3.2 Mesh current analysis

3.3 Bipolar junction transistors (BJTs)

节点电压法 (Nodal voltage analysis)

- 不含电压源和受控源

选择参考节点（一般选择接地的节点），参考节点电位为0，其余 $(n-1)$ 个非参考节点电压分别为 $V_1, V_2, \dots, V_{n-1}$ ，均为相对于参考节点的电位

2. 对 $(n-1)$ 个非参考节点用KCL列方程，用欧姆定律节点电压表示各支路电流
3. 求解方程组

注意：

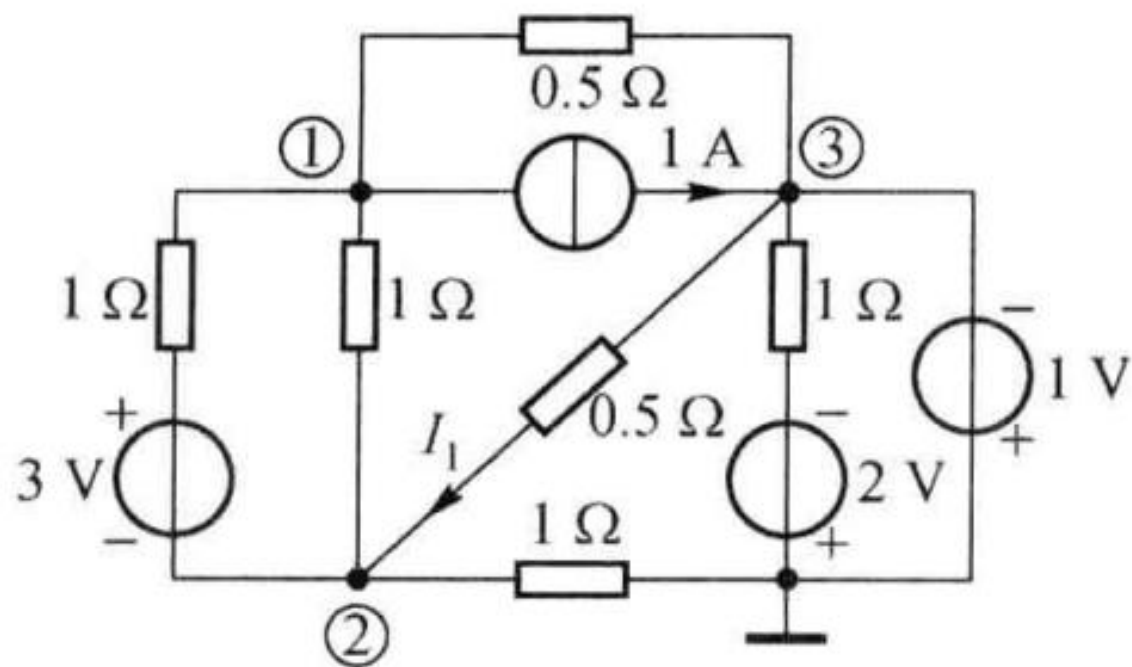
1. 通过电阻的电流总是从高电位流向低电位
2. 除了电流源支路外，其余电流方向可以任意标记

- 含电压源，独立源或受控源

1. 如果电压源接在参考节点与非参考节点之间，非参考节点电压就等于电压源电压
2. 如果电压源接在两个非参考节点之间，这两个非参考节点构成超节点（supernode）：对整个超节点列KCL，对超节点内部列KVL，对其余节点列KCL

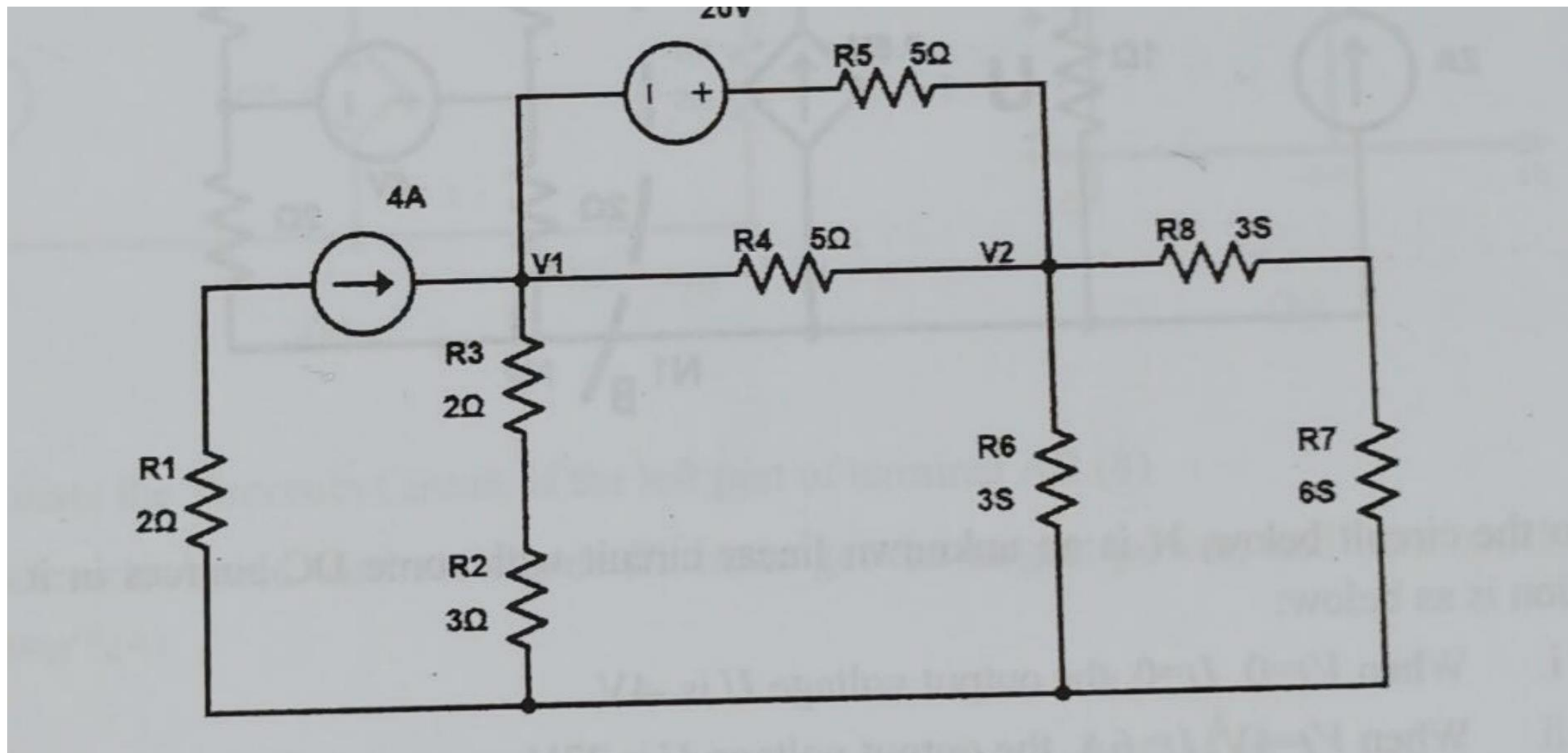
超节点由两个非参考节点和其间电压源以及与之并联的元件组成

(其他教材) 求 I_1



图题 2.25

典例讲解：求V1和V2



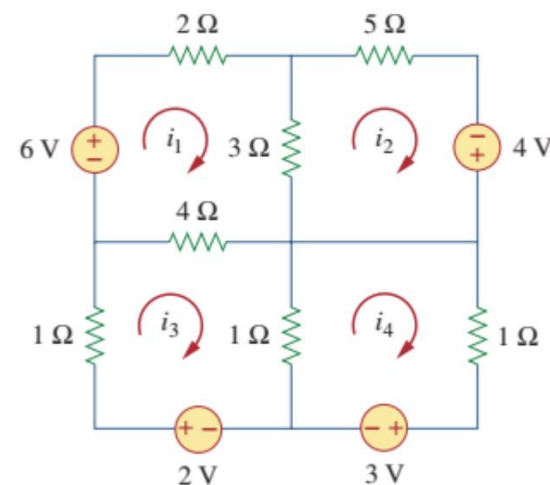
网孔电流法(Mesh analysis)

- 网孔分析法 **Mesh Analysis** (不含电流源) :

1. 为 n 个网孔分别指定网孔电流 $i_1, i_2, \dots, i_n$
2. 对每个网孔列KVL, 根据欧姆定律用网孔电流表示各个电压
3. 联立求解

注意:

1. 网孔电流方向是任意假设的
2. 相邻网孔的支路的电流不等于网孔电流!



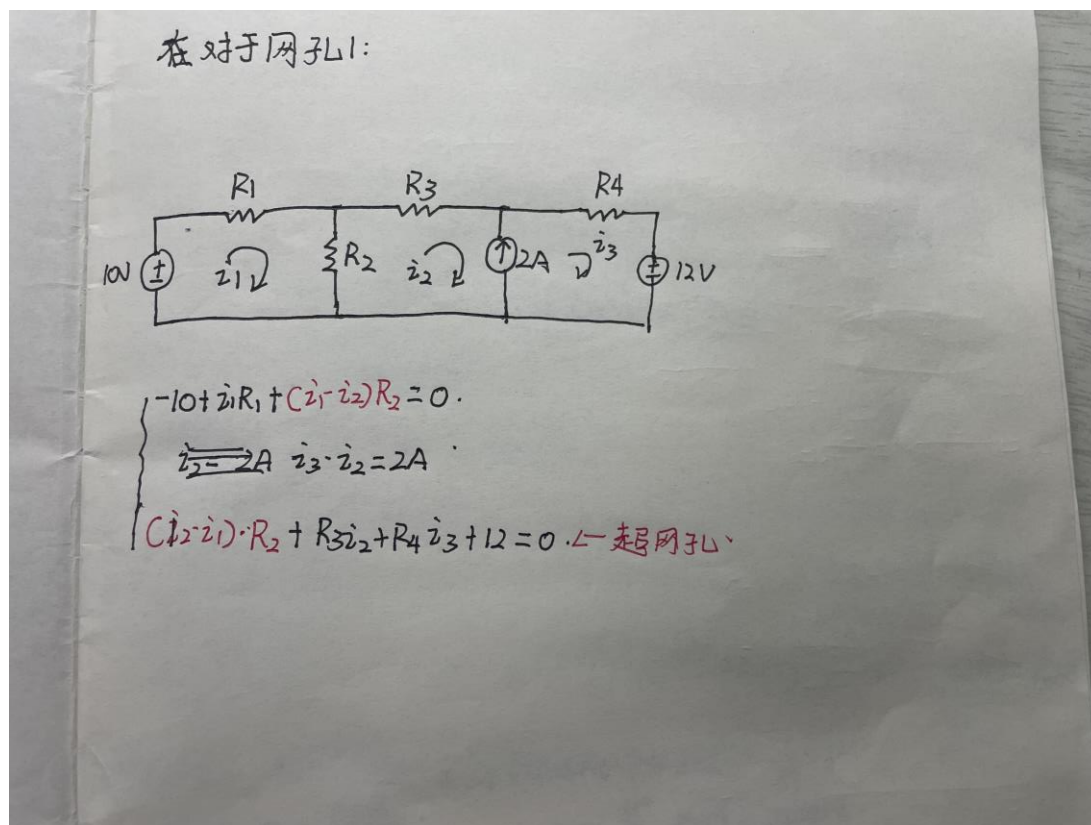
- 网孔分析法 (含电流源 (独立源或受控源)) :

1. 若电流源仅存在于一个网孔中, 这个网孔电流等于电流源电流
2. 若两个网孔共用一个电流源, 产生一个超网孔: 对超网孔列KVL, 对两个网孔共有支路上的节点列KCL

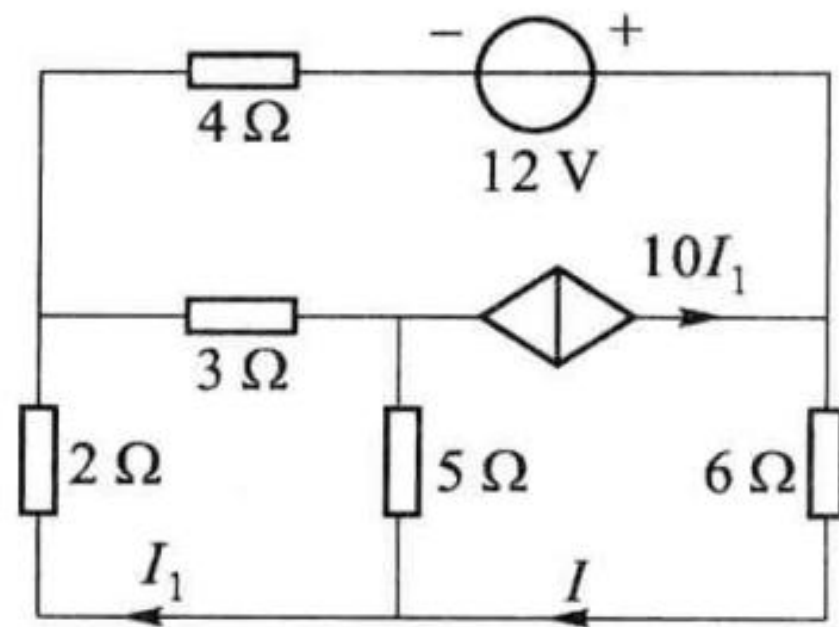
超网孔由这两个网孔并且去掉电流源和与之串联的元件组成

易错点分析:

1. 统一方向: 统一顺时针方向设电流。
2. 对于横跨两个网孔的电阻, 当以该网孔为分析对象时, 其所代表的电流为正。

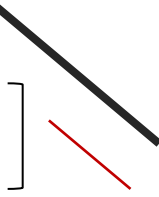


(其他教材) 典例讲解: 求 I_1

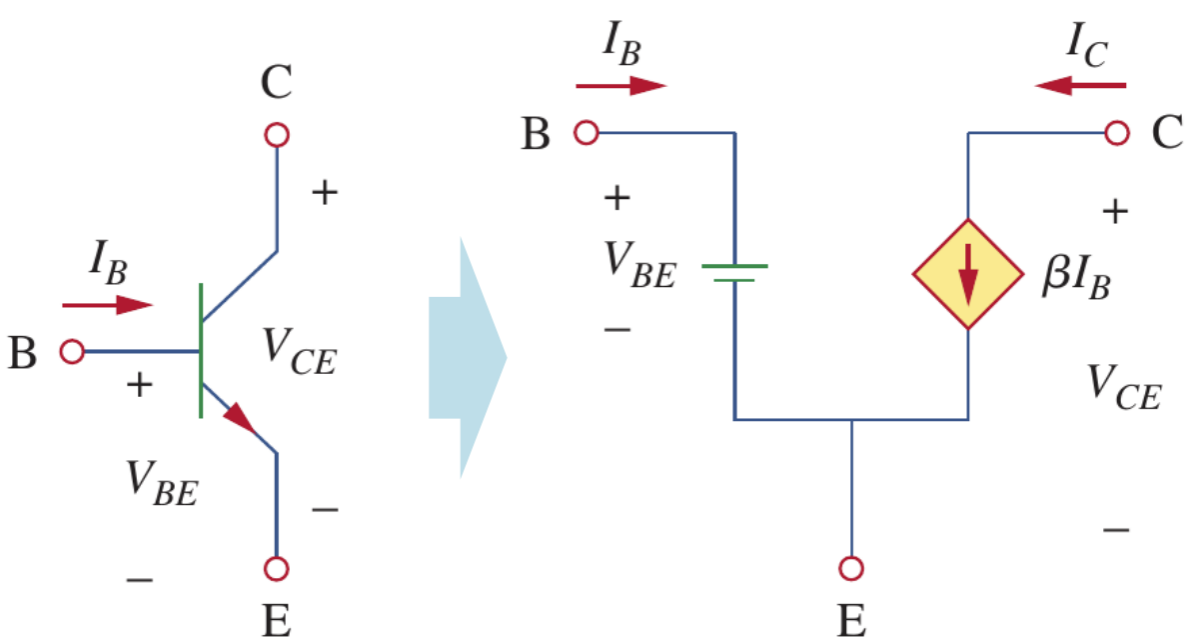
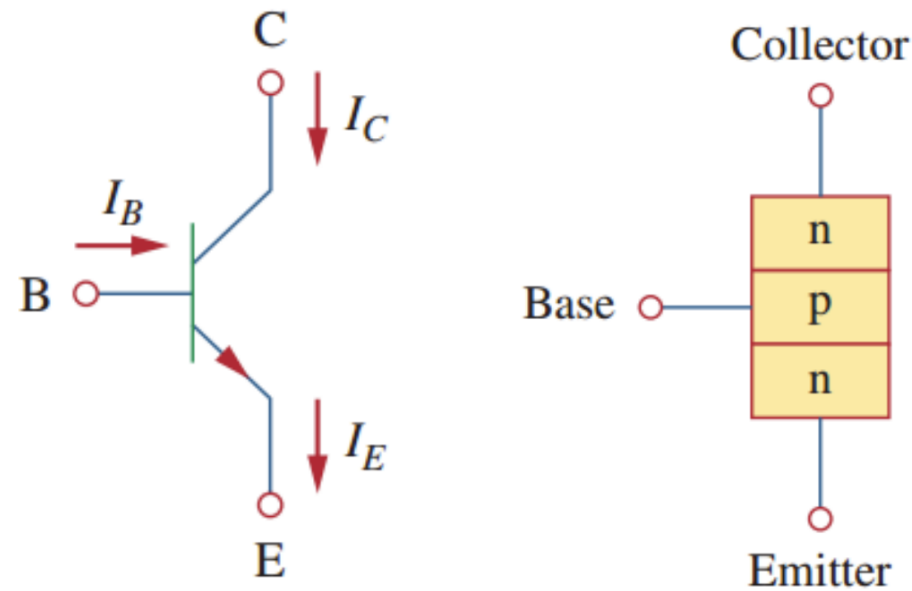


图题 2.14

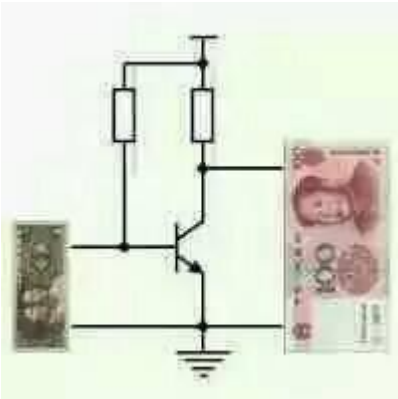
选哪种方法？



3.3 BJT (Bipolar junction transistors)



$$I_E = I_B + I_C, I_E = (1 + \beta) \cdot I_B$$



作业题

3.89 For the transistor circuit shown in Fig. 3.125, find I_B and V_{CE} . Let $\beta = 100$ and $V_{BE} = 0.7\text{ V}$. (10')

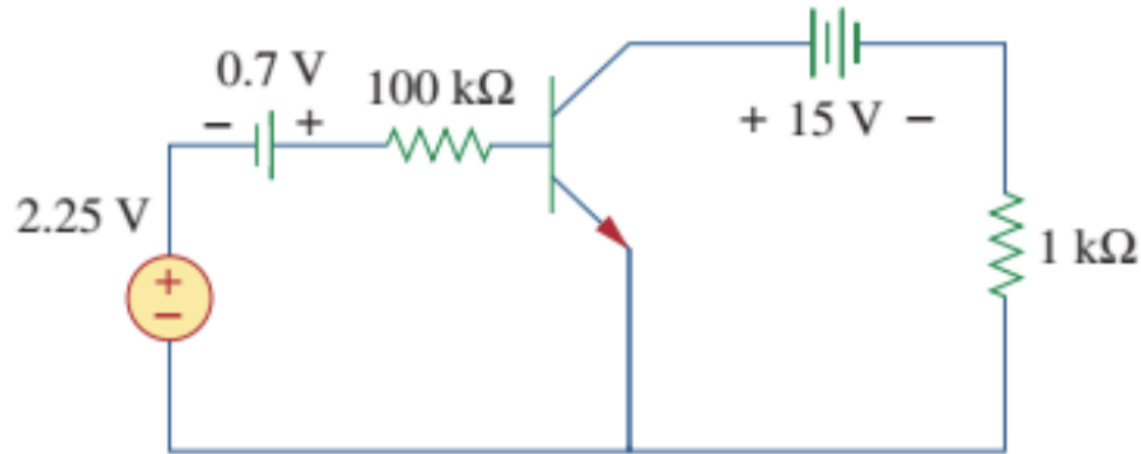
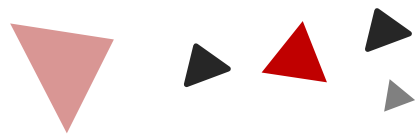


Figure 3.125

电路分析方法

4



Chapter

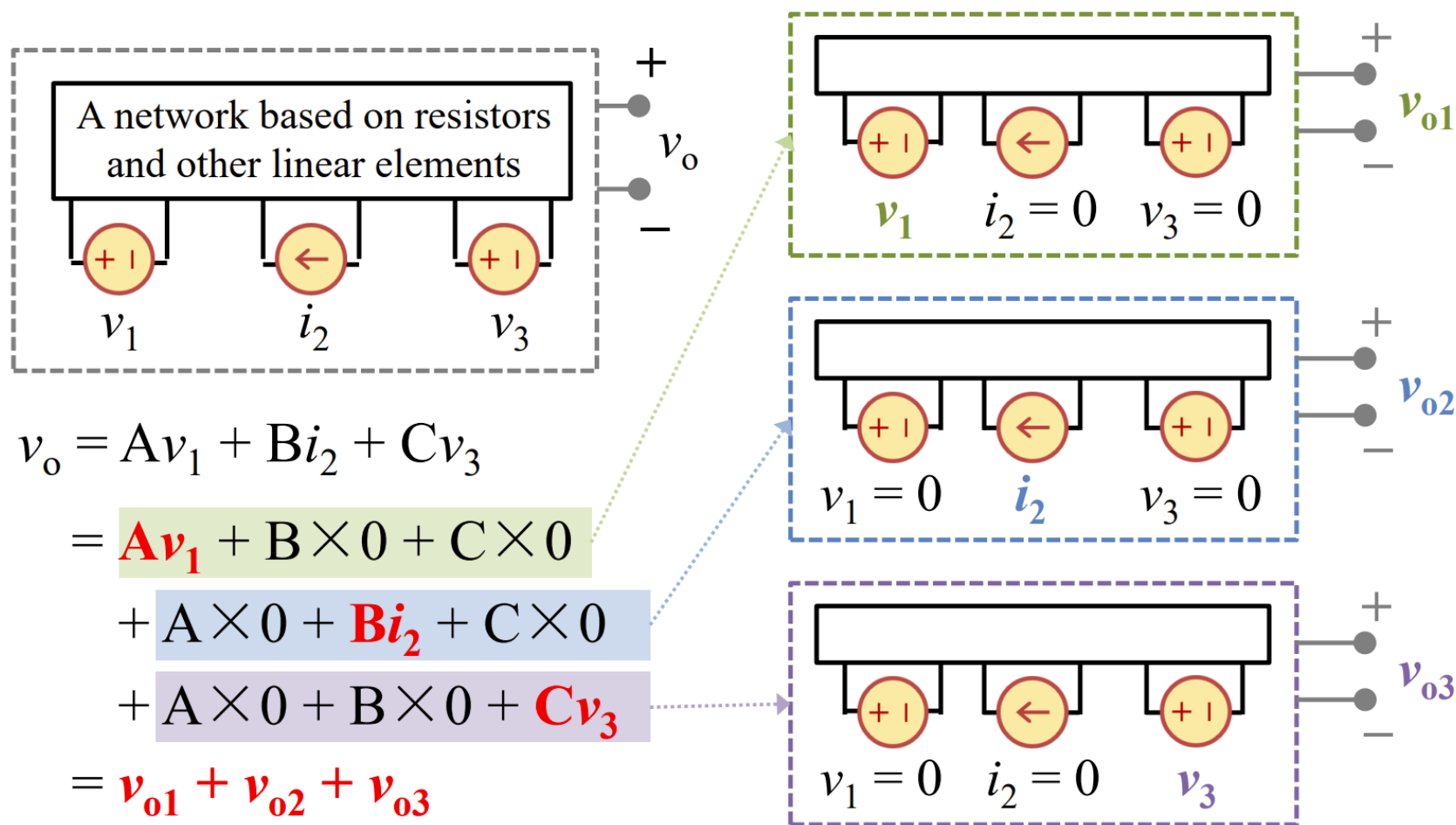
4.1 Superposition Theorem 叠加定理

4.2 Source Transformation 源变换

4.3 Thevenin's theorem and Norton's theorem 戴维南定理和诺顿定理

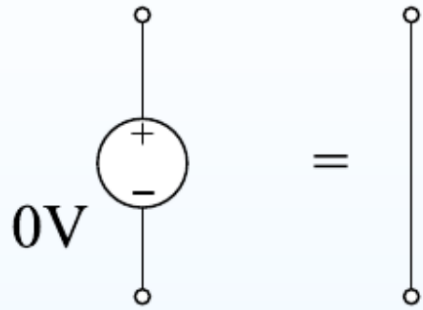
4.4 Maximum power transfer theorem 最大功率传输定理

4.1 叠加定理 (Superposition Theorem)

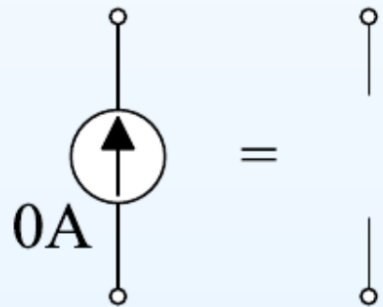


4.1 叠加定理 (Superposition Theorem)

- Zero-value sources (How to turn off independent sources)

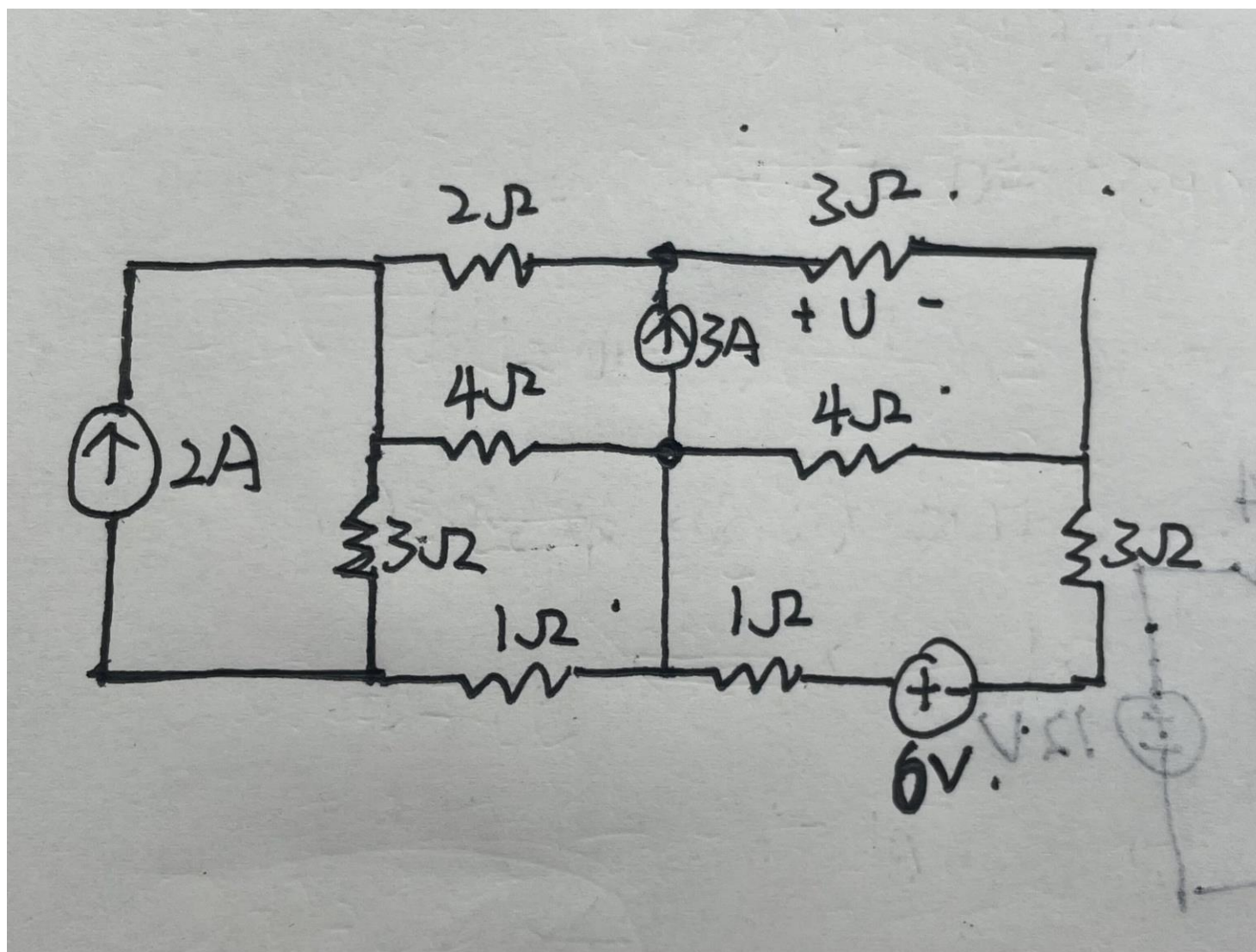


A zero-valued voltage source has zero volts between its terminals for any current. It is equivalent to a short-circuit or piece of wire or resistor of $0\ \Omega$ (or $\infty\ \text{S}$).



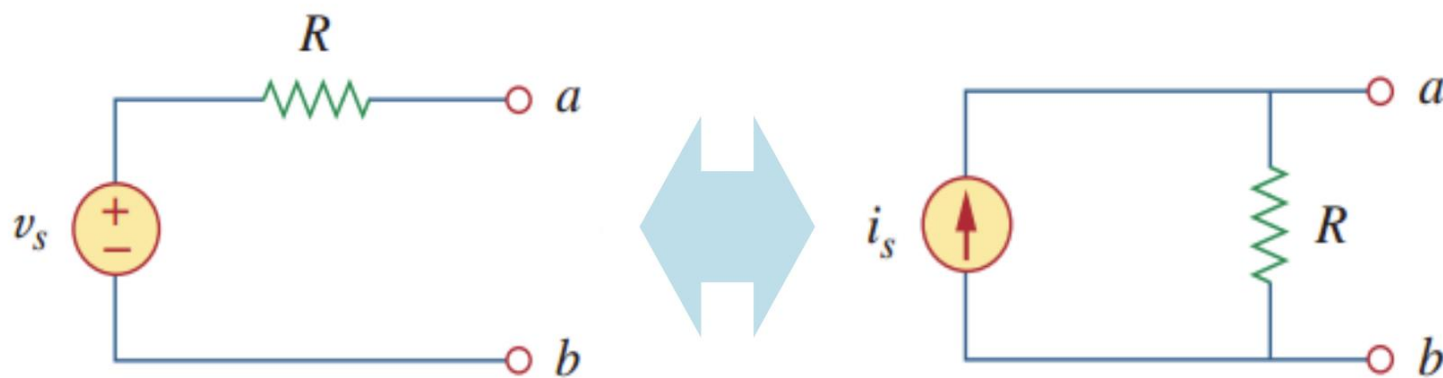
A zero-valued current source has no current flowing between its terminals. It is equivalent to an open-circuit or a broken wire or a resistor of $\infty\ \Omega$ (or $0\ \text{S}$).

(2021期中) 典例讲解: 利用叠加定理求U



4.2 源变换 (Source Transformation)

$$a) \quad R = R' \text{ \& } v_s = R \cdot i_s$$



源变换YYDS!

电路化简哪家强?

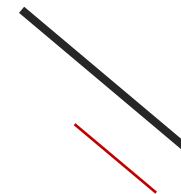
减少网孔数量 网孔电流法

方程未知数数量减少

减少节点数量 节点电压法

下笔前先观察是否能源变换

源变换注意事项



4.3 戴维南定理和诺顿定理

戴维南定理

V_{Th}: 端口开路电压

R_{Th}: 独立源关闭时端口的输入电阻

求R_{Th}:

(1) 当网络中不包含受控源时, 关闭所有独立源,
R_{Th}就是从**ab**两端向网络看进去的输入电阻

(2) 当网络中包含受控源时, 关闭独立源,
选择一下两种方法任意一种:

1. 在**ab**端外加一个电压源**V₀**, 计算端口电流**i₀**
2. 在**ab**端外加一个电流源**i₀**, 计算端口电压**V₀**

Norton's Theorem 诺顿定理

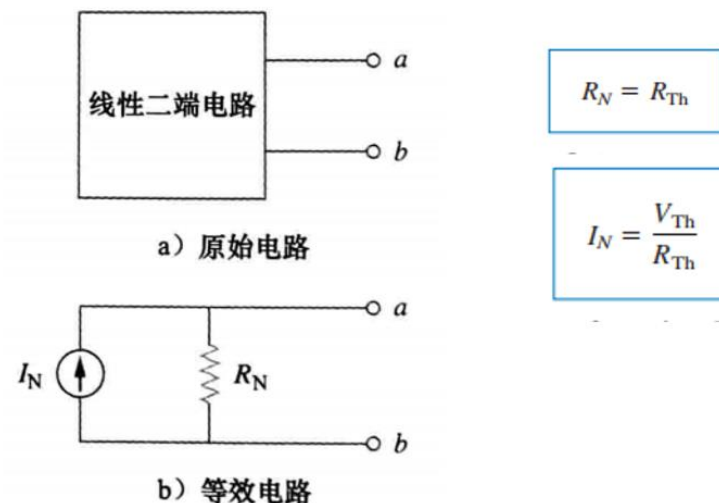
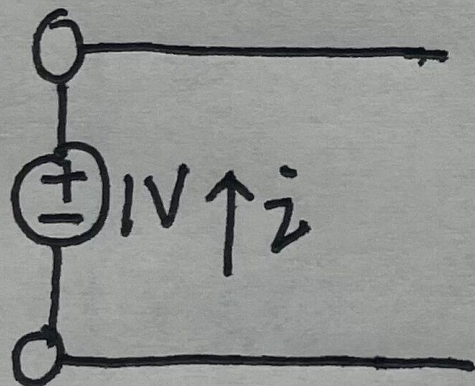
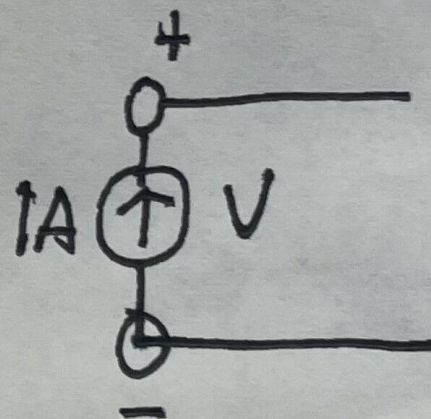


图 4-37 诺顿等效电路

易错提示：求等效电阻时的方向问题

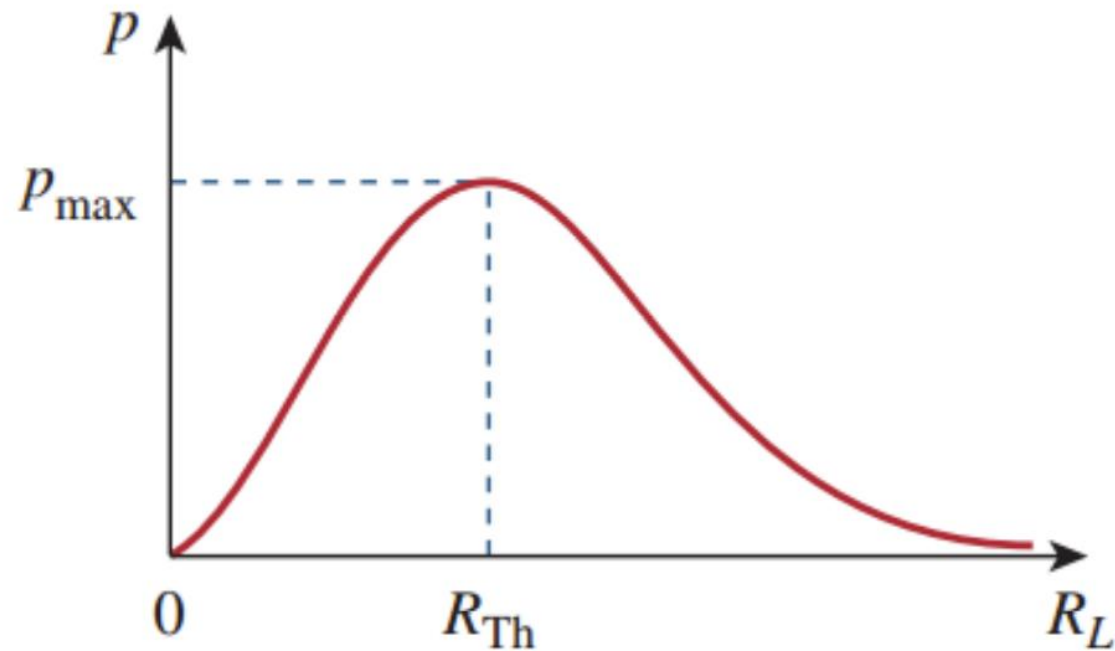


$$R = \frac{1}{z}$$



$$R = \frac{V}{1}$$

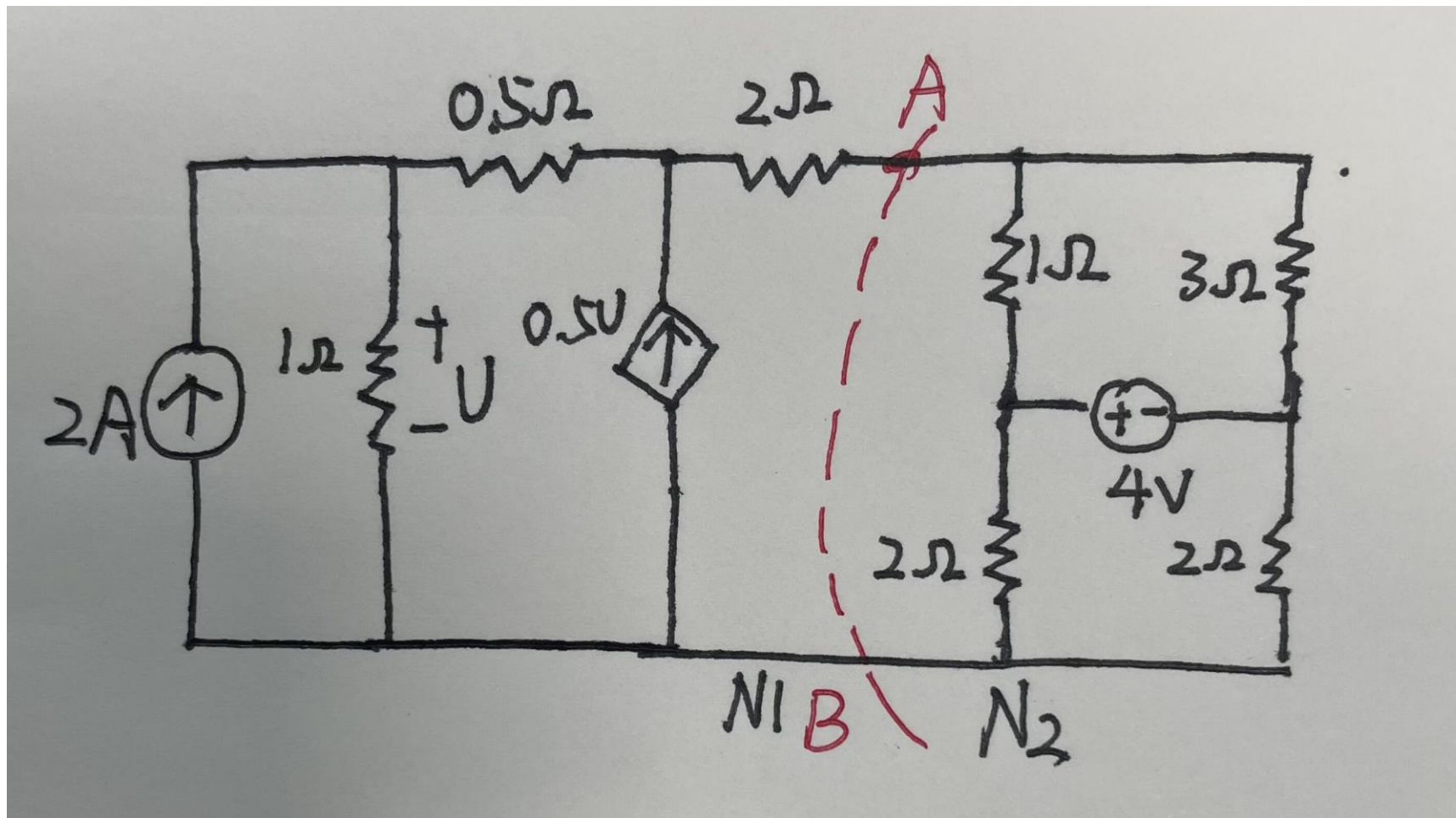
4.4 最大功率传输定理



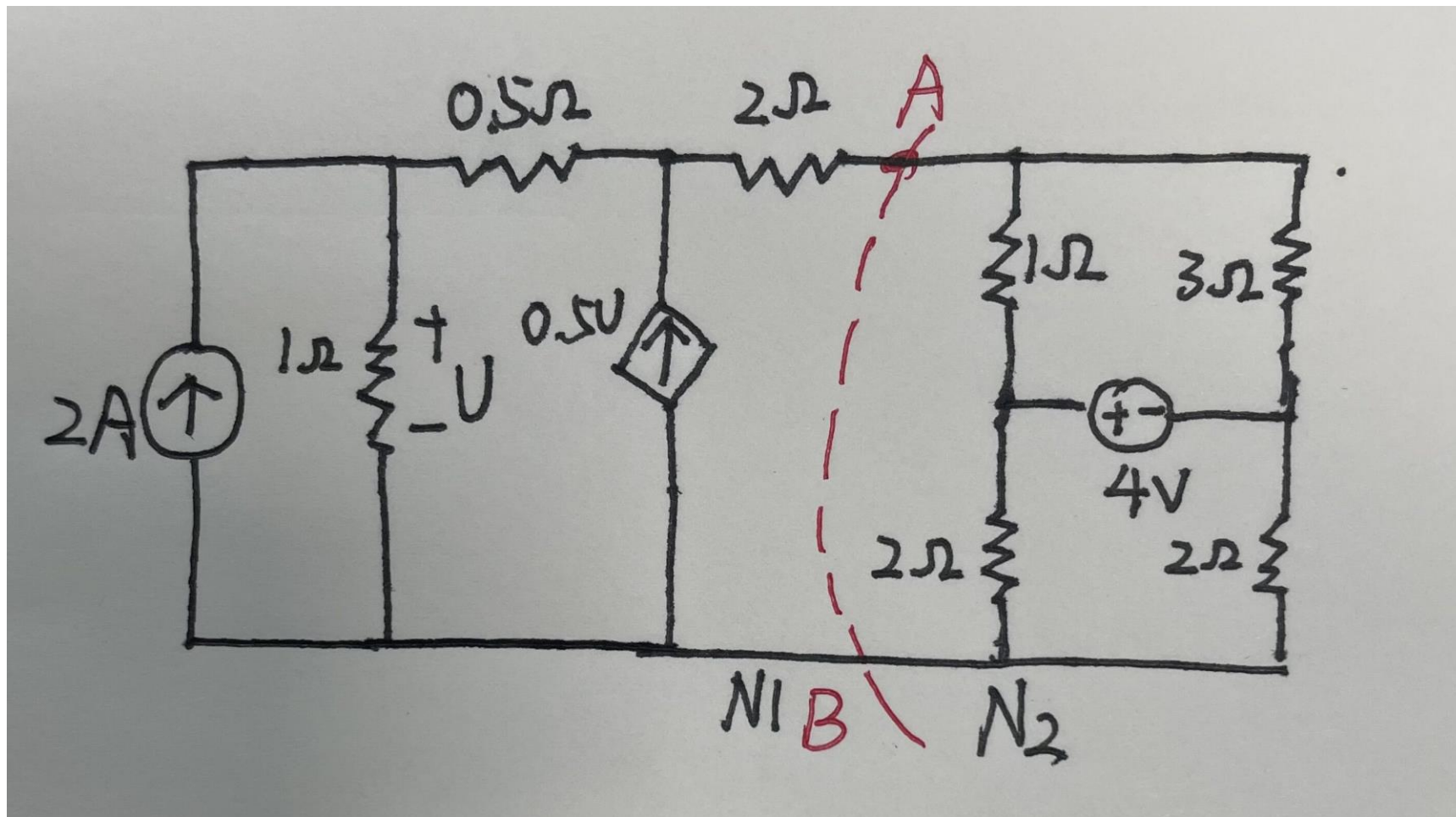
$$R_L = R_{Th} \quad p_{max} = \frac{V_{Th}^2}{4R_{Th}}$$

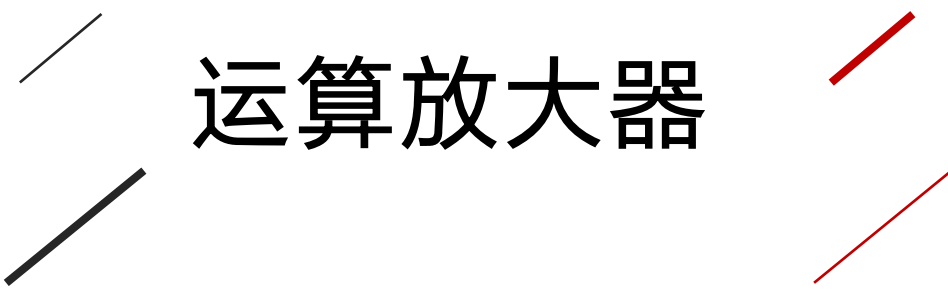
R_{th} : 从负载电阻看进去的戴维南等效电阻

- (2021期中) (1) 求 N_1, N_2 的戴维南等效电路 (A, B分割)
(2) 在A, B之间接一个电阻, 求解其最大功率和达到最大功率时的电阻值



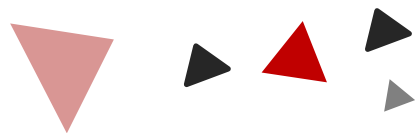
- (2021期中) (1) 求AB左侧的戴维南等效
(2) 在A,B之间接一个电阻, 求解其最大功率和达到最大功率时的电阻值





运算放大器

5



Chapter

5.1 黄金法则

5.2 Inverting and noninverting amplifiers

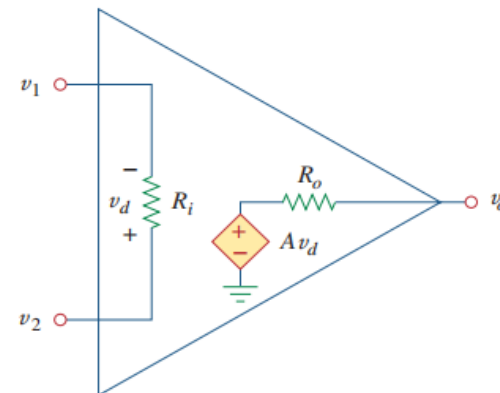
反相器/非反相器

5.3 Difference amplifiers and Summing amplifiers

差分放大器/加法器

运算放大器:

- 运算放大器本质是电子元件
- 其对信号进行放大，相加等
- 需要额外的电源（题目中一般不会画出）



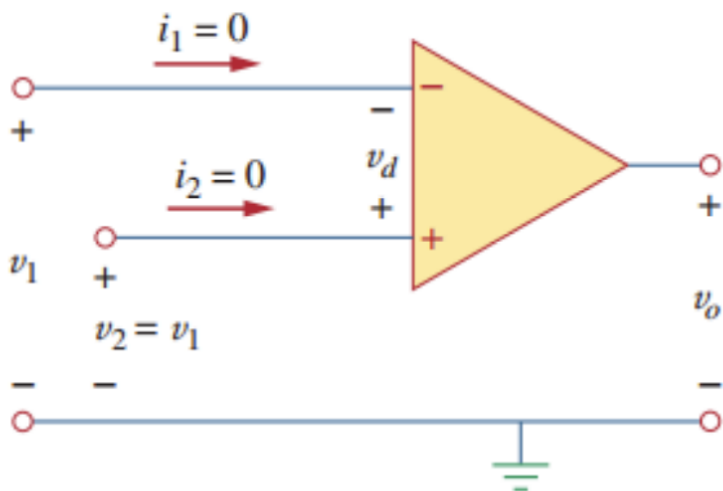
引入负反馈:

- 运放的开环增益 A 很大，接近于无穷，如果不加入负反馈，直接使用开环增益，即使两个输入端之间加微小电压，输出电压都将超过其线性范围，工作在非线性区。
- 只有引入负反馈，才能保证运放工作在线性区，实现一定的功能，此时运放为闭环增益，闭环增益只与运放外部连接元件有关，与开环增益 A 无关

5.1 黄金法则

- THE GOLDEN RULES:

Ideal op-amps	A	R_i	R_o
	∞	∞	0



Golden rules for analyzing ideal op-amp:

(1) Currents into both input terminals are 0:

$$i_1 = i_2 = 0$$

(2) The voltage across the input terminals is 0:

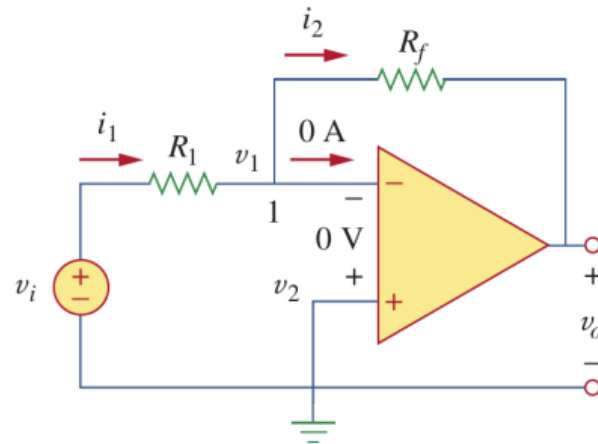
$$v_2 = v_1$$

- To calculate voltages, the two input terminals can be considered as “shorted”. ($R_i \rightarrow \infty$, 虚短)
- To calculate currents, the two input terminals and the internal portion of the op-amp can be considered as “open”. ($v_2 - v_1 = v_o/A$, $A \rightarrow \infty$, $v_2 - v_1 = 0$, 虚断。

推荐方法：节点电压法

5.2 反相器和非反相器

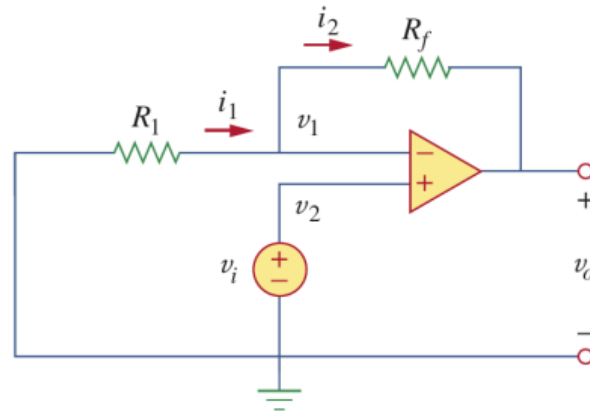
Inverting
op-amp:



Negative

$$v_o = -\frac{R_f}{R_1} v_i$$

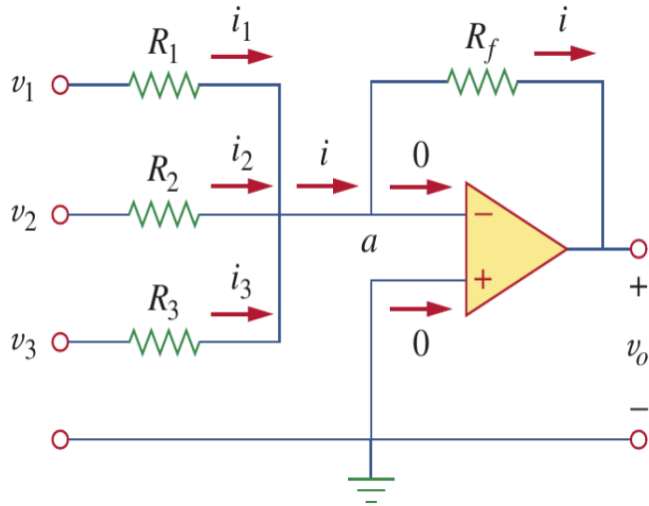
Noninverting
op-amp:



Positive

$$v_o = \left(1 + \frac{R_f}{R_1}\right) v_i$$

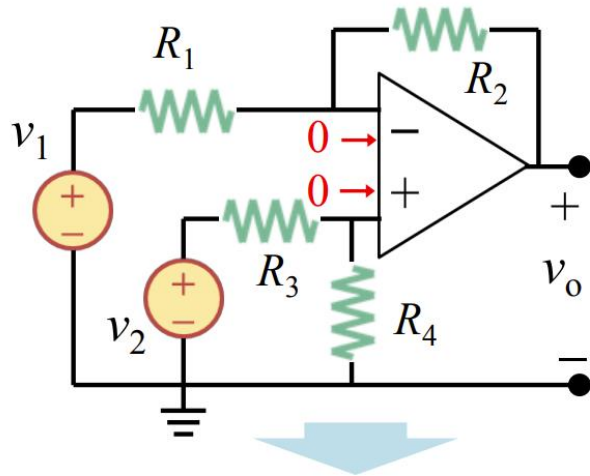
5.3 Difference amplifiers and Summing amplifiers



$$v_o = -\frac{R_f}{R_1}v_1 - \frac{R_f}{R_2}v_2 - \frac{R_f}{R_3}v_3$$

加法器

怎么算？叠加定理

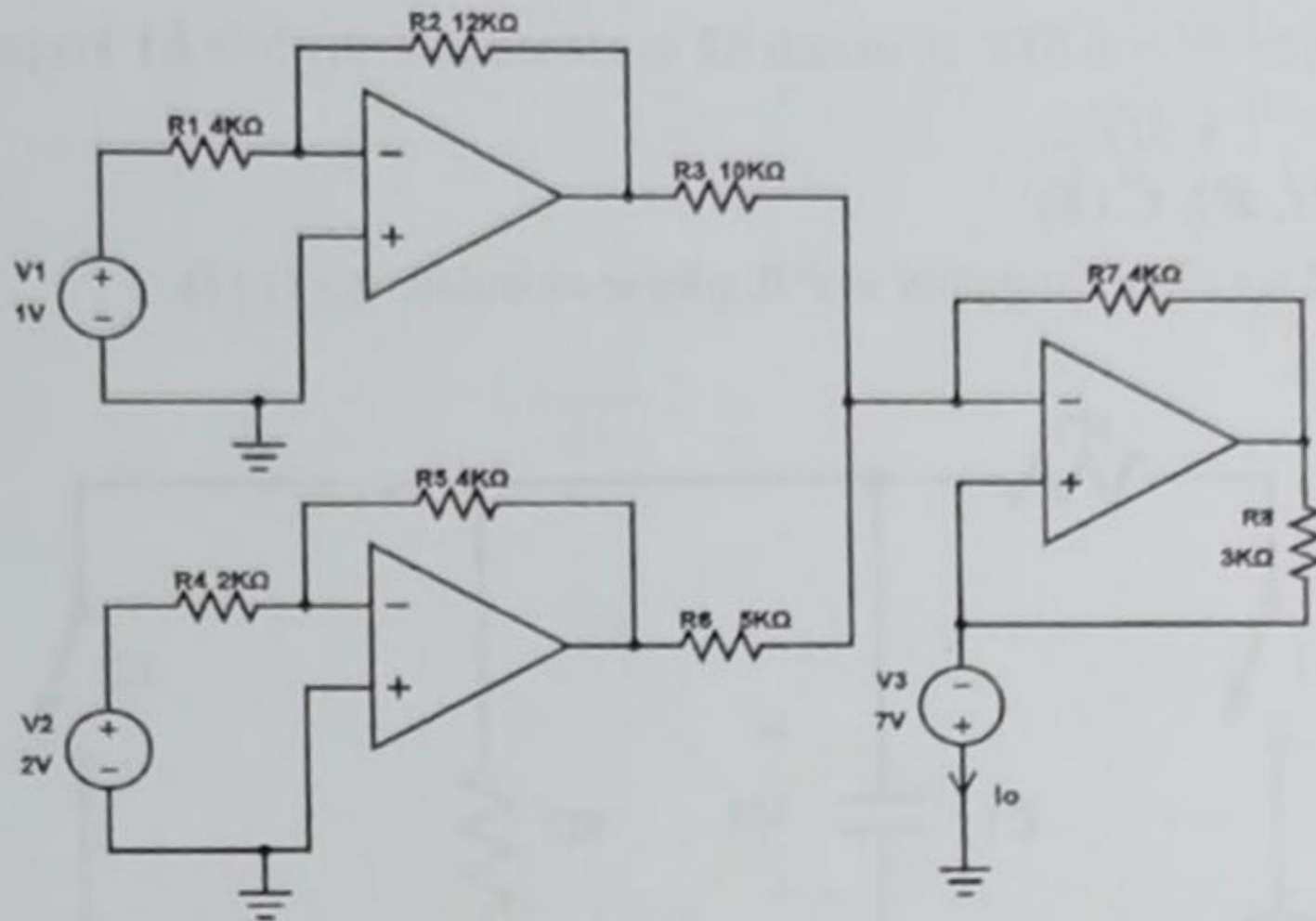


$$v_o = \frac{R_2(1 + R_1/R_2)}{R_1(1 + R_3/R_4)}v_2 - \frac{R_2}{R_1}v_1$$

$$\frac{R_1}{R_2} = \frac{R_3}{R_4} \quad v_o = \frac{R_2}{R_1}(v_2 - v_1)$$

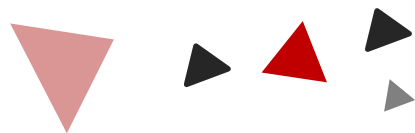
差分放大器

(2021期中) 求 I_1



电容和电导

6



Chapter

6.1 Capacitors

6.2 Inductors

6.3 Integrators and differentiators 积分器和微分器

6.1 电容 (Capacitors)

几个重要的关系:

$$C = \frac{\epsilon A}{d}$$

$$q = Cv$$

$$i = C \frac{dv}{dt}$$

$$v(t) = \frac{1}{C} \int_{-\infty}^t i(\tau) d\tau = \frac{1}{C} \int_{t_0}^t i(\tau) d\tau + v(t_0)$$

$$W = \frac{1}{2} C V^2 = q^2 / 2C$$

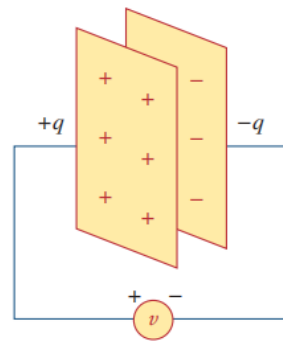
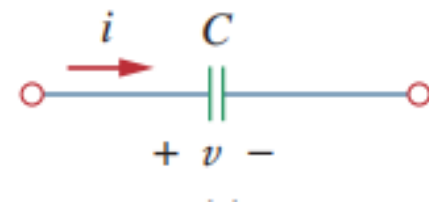
并联:

$$C_{eq} = C_1 + C_2 + \dots + C_n$$

串联:

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_n}$$

对于直流电路: 开路
电容上电压不会突变



6.2 电感 (Inductors)

几个重要的关系:

$$L = \frac{N^2 \mu A}{l}$$

$$v = L \frac{di}{dt}$$

$$p = vi = \left(L \frac{di}{dt} \right) i = iL \frac{di}{dt}$$

$$i(t) = \frac{1}{L} \int_{-\infty}^t v(\tau) d\tau = \frac{1}{L} \int_{t_0}^t v(\tau) d\tau + i(t_0)$$

$$W = \frac{1}{2} Li^2$$

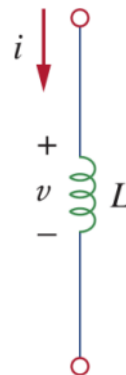
串联:

$$L_{eq} = L_1 + L_2 + \dots + L_n$$

并联:

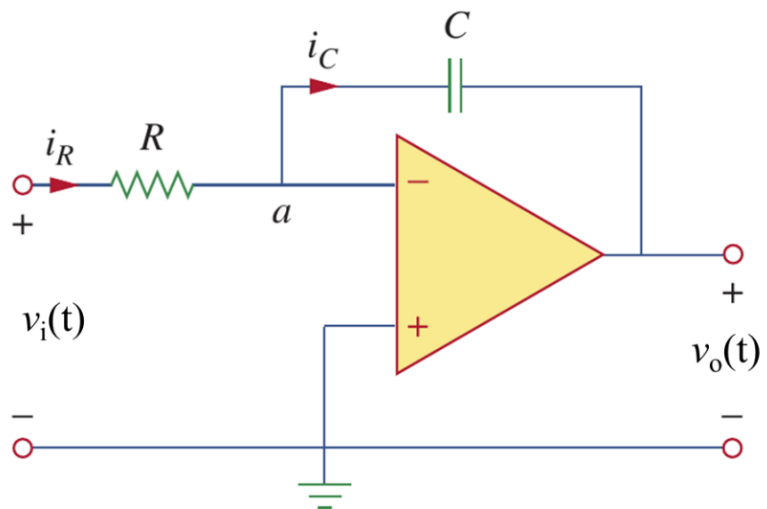
$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \dots + \frac{1}{L_n}$$

对于直流电路是导线
其电流不能突变



*Air-core
inductors*

6.3 积分器和微分器

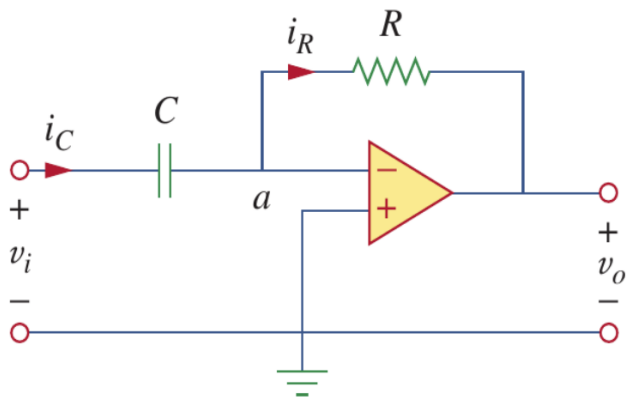


积分器：电容位于反馈通路上

$$(v_i - 0)/R = i_c = C \frac{d(0 - v_o)}{dt}$$

$$-(v_i)/R dt = C dv$$

$$v = -\frac{1}{RC} \int_{t_0}^t v_i dt + v(t_0)$$



微分器：电容位于放大通路上

$$\frac{Cd(v_i - 0)}{dt} = \frac{0 - v_o}{R}$$

$$v_o = -RC \frac{dv_i}{dt}$$

作业题

Fig. 6

7. (15') The inductors in Fig.7 are initially charged and are connected to the black box at $t = 0$. If $i_1(0) = 4$ A, $i_2(0) = -2$ A, and $v(t) = 50e^{-200t}$ mV, $t \geq 0$, find:
- (a) the energy initially stored in each inductor,
 - (b) the total energy delivered to the black box from $t = 0$ to $t = \infty$,
 - (c) $i_1(t)$ and $i_2(t)$, $t \geq 0$,
 - (d) $i(t)$, $t \geq 0$.

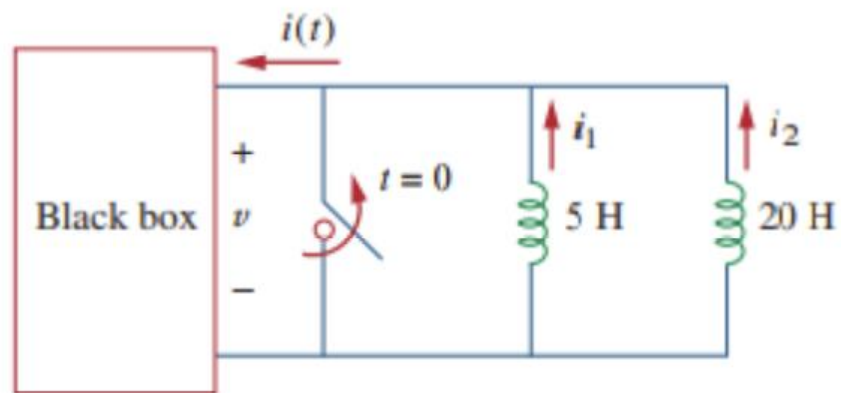
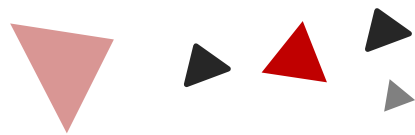


Fig. 7

一阶电路

7



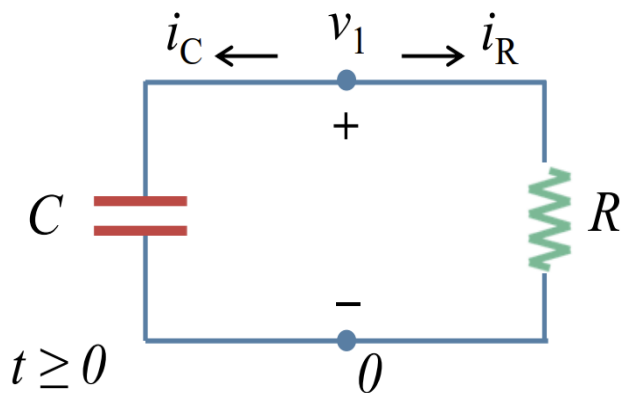
Chapter

7.1 Natural response of RC and RL circuits

7.2 Step-response of RC and RL circuits

7.1 自由响应RC (Natural response of RC and RL circuits)

初始条件激发RC电路
无源RC



$$i_R(t) = \frac{v(t)}{R} = \frac{V_0}{R} e^{-t/\tau}$$

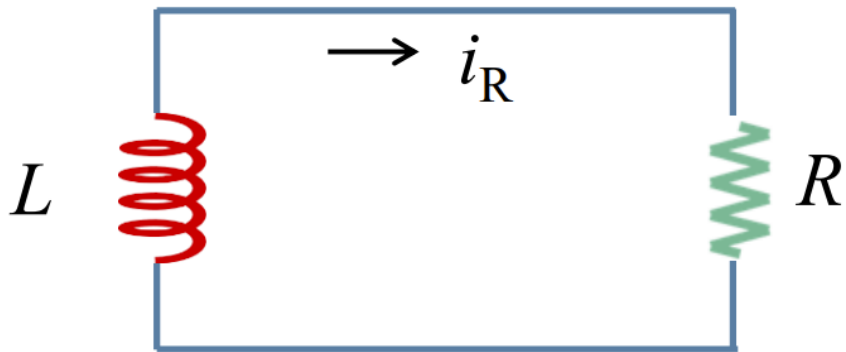
$$p(t) = v i_R = \frac{V_0^2}{R} e^{-2t/\tau}$$

$$w_R(t) = \int_0^t p(\lambda) d\lambda = \frac{V_0^2}{R} \int_0^t e^{-2\lambda/\tau} d\lambda = -\frac{1}{2} \tau \cdot \frac{V_0^2}{R} e^{-2\lambda/\tau} \Big|_0^t = \frac{1}{2} C V_0^2 (1 - e^{-2t/\tau})$$

$$v(t) = V_0 e^{-t/RC} = V_0 e^{-t/\tau}$$

7.1 自由响应RL (Natural response of RC and RL circuits)

初始条件激发RL电路
无源RL



$$v_R(t) = i(t)R = I_0 R e^{-t/\tau}$$

$$i(t) = i(0)e^{-Rt/L} = i(0)e^{-t/\tau}$$

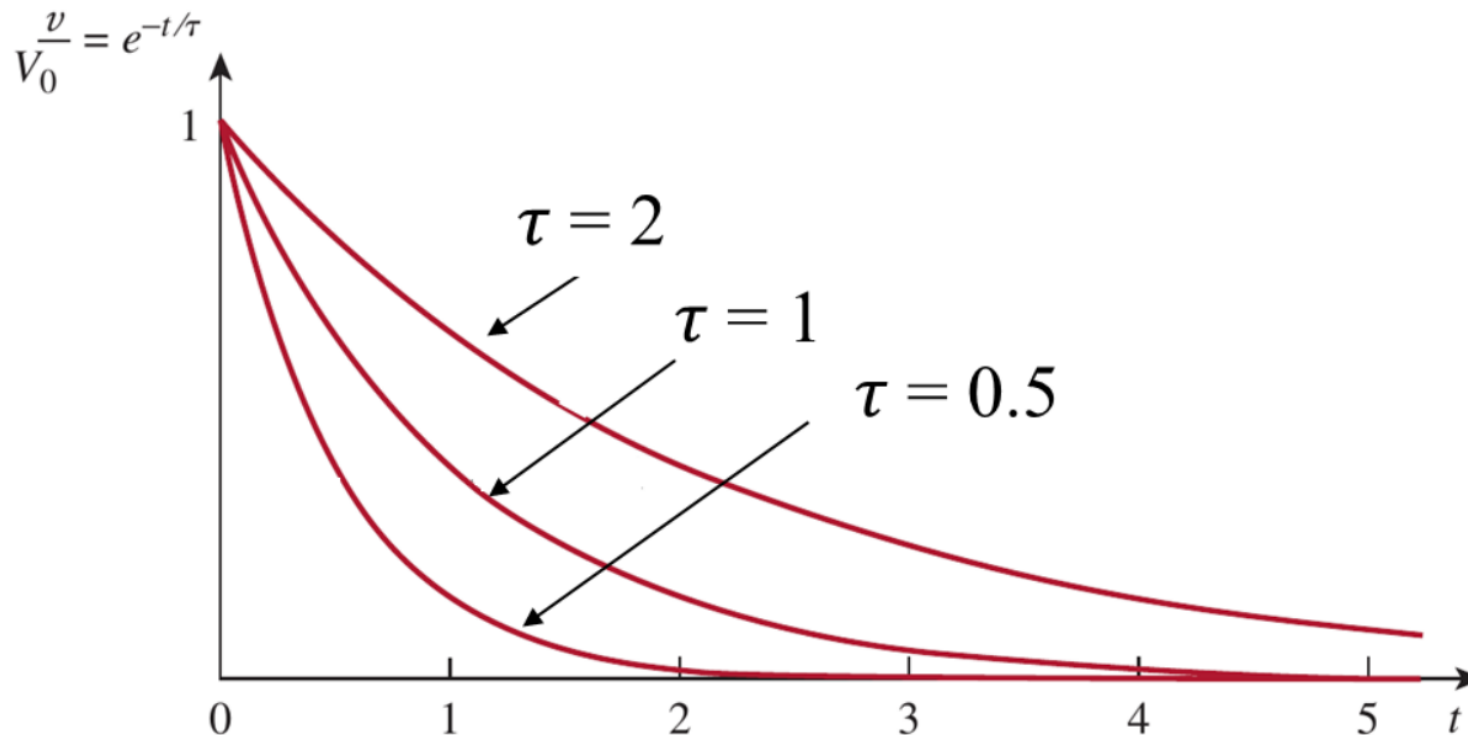
$$p(t) = v_R i = I_0^2 R e^{-2t/\tau}$$

$$w_R(t) = \int_0^t p(\lambda) d\lambda = I_0^2 R \int_0^t e^{-2\lambda/\tau} d\lambda = -\frac{1}{2}\tau \cdot I_0^2 R e^{-2\lambda/\tau} \Big|_0^t = \frac{1}{2} L I_0^2 (1 - e^{-2t/\tau})$$

7.1 时间常数 (time constant)

$$\tau = RC$$

$$\tau = L/R$$



7.2阶跃响应 (Step-response of RC and RL circuits)

阶跃响应分析方法:

1. 自由响应+受迫响应 (**forced response**)

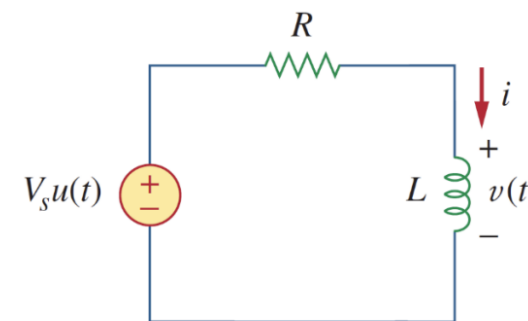
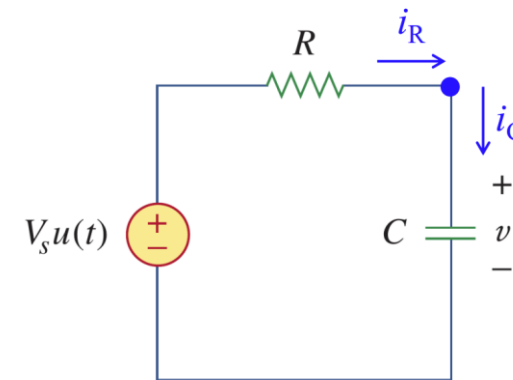
$$v(t) = v_n + v_f = v(0)e^{-t/\tau} + v(\infty)[1 - e^{-t/\tau}]$$

$$i(t) = i_n + i_f = i(0)e^{-t/\tau} + i(\infty)[1 - e^{-t/\tau}]$$

2. 瞬态响应+暂态响应(**transient response + steady-state response**)

$$v(t) = v_{ss} + v_t = v(\infty) + [v(0) - v(\infty)] e^{-t/\tau}$$

$$i(t) = i_{ss} + i_t = i(\infty) + [i(0) - i(\infty)] e^{-t/\tau}$$



作业题

3. (10') In the circuit of Fig. 3, find $i(t)$ for $t > 0$ if $i(0) = 6$ A.

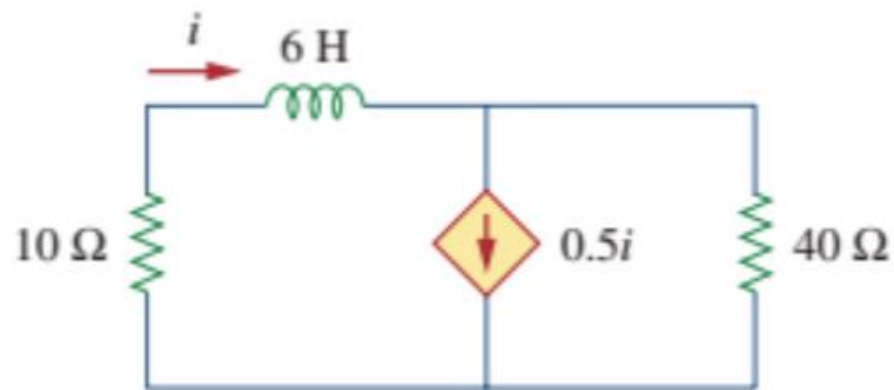
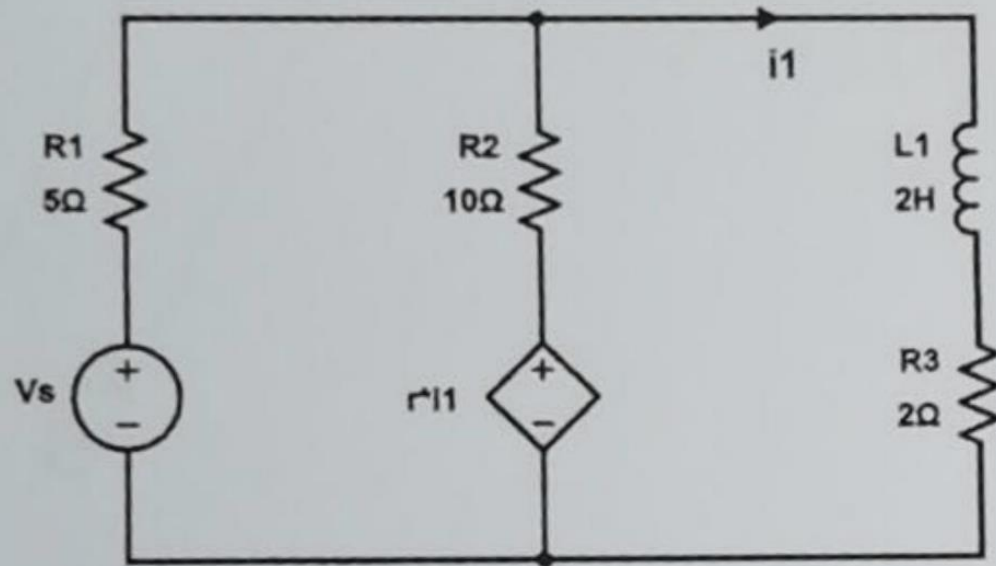
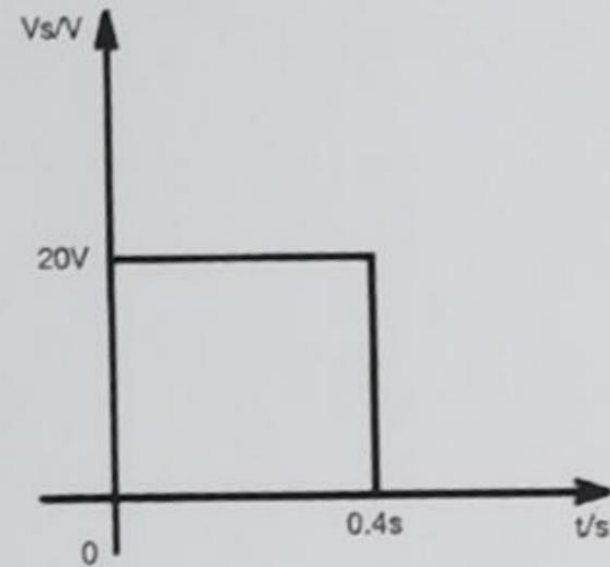


Fig. 3

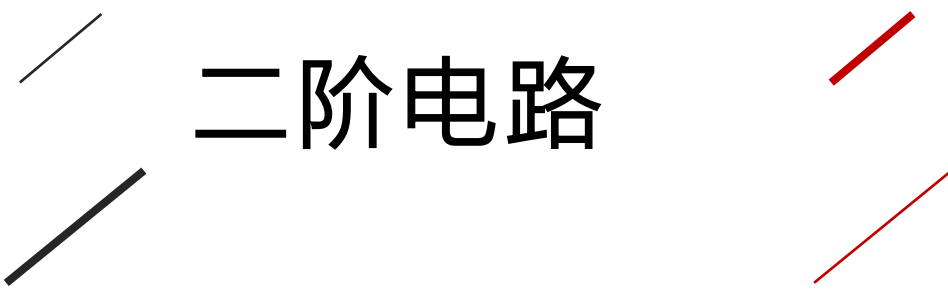
(2021期中) $i_1 = 2\text{A}$. $R = 1\Omega$, 求 $i_1(t)$



(a)

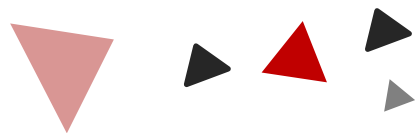


(b)

The title is flanked by two sets of parallel diagonal lines. On the left, two black lines of different lengths. On the right, two red lines of different lengths.

二阶电路

8



8.1 Second Order Circuits

8.2 Natural response of RLC circuits

8.3 Step response of RLC circuits

补充知识：二阶常系数线性微分方程

标准形式： $y'' + py' + qy = 0$

特征方程： $r^2 + pr + q = 0$

方程的通解形式：

(1) 有两个实根： r_1, r_2 $y = C_1 e^{r_1 t} + C_2 e^{r_2 t}$

(2) 有两个相等的实根： r $y = (C_1 + r C_2) * e^{rt}$

(3) 有两个复根： $\alpha + i\beta, \alpha - i\beta$

$$y = e^{\alpha} (C_1 \cos(\beta x) + C_2 \sin(\beta x))$$

8.2 RLC电路的自然响应 (Natural response of RLC circuits)

$$s_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}} \quad s_2 = -\frac{R}{2L} - \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}, \quad s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$$

$$\alpha = \frac{R}{2L}, \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

ω_0 : resonant frequency, 谐振频率

α : neper frequency , damping factor 奈培频率,
阻尼系数

8.2 RLC电路的自然响应 (Natural response of RLC circuits)

过阻尼(overdamped): $\alpha > \omega_0$:

$$y = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

临界阻尼(critically damped): $\alpha = \omega_0$:

$$y = (C_1 + r C_2) * e^{-\alpha t}$$

欠阻尼(underdamped) : $\alpha < \omega_0$:

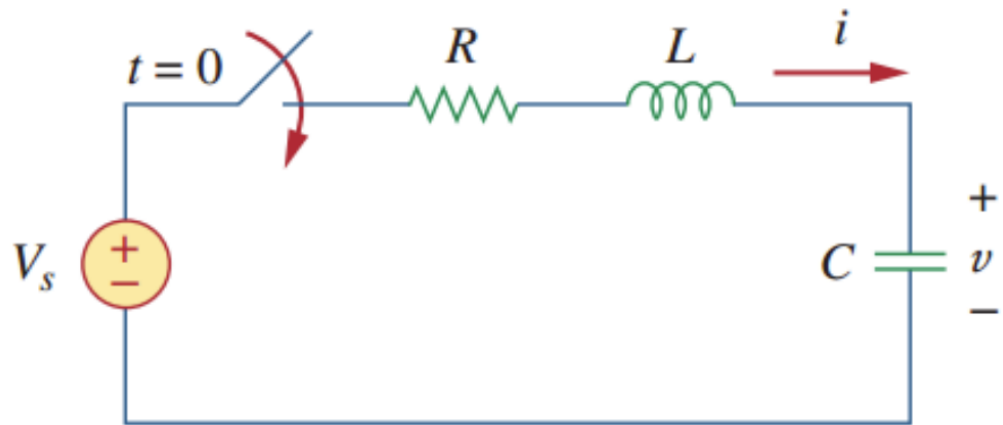
$$y = e^{-\alpha} (A_1 \cos(w d x) + A_2 \sin(w d x)) \quad w d = \sqrt{w_0^2 - \alpha^2}$$

无阻尼振荡(undamped): $\alpha = 0$

$$y = (A_1 \cos(w_0 x) + A_2 \sin(w_0 x))$$

8.3 RLC电路的阶跃响应 (Step response of RLC circuits)

串联RLC电路:



8.3 RLC电路的阶跃响应 (Step response of RLC circuits)

- Total step response of parallel RLC circuits

Overdamped: $i(t) = I_s + A_1 e^{s_1 t} + A_2 e^{s_2 t}$

Critically damped: $i(t) = I_s + (A_1 + A_2 t) e^{-\alpha t}$

Underdamped: $i(t) = I_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t}$

$$i(t) = i_{ss}(t) + i_t(t)$$

**Complete
response**

=

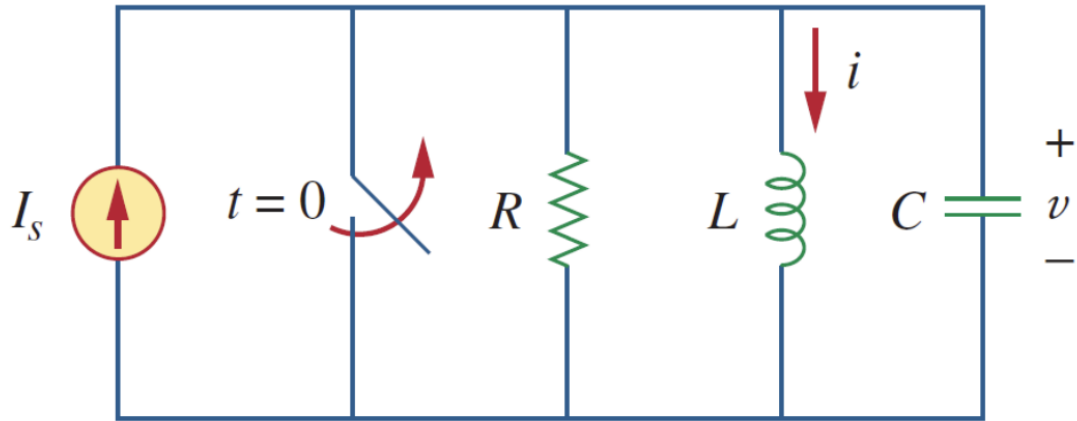
Steady-state response
Permanent part $i(\infty)$
After all transients are done

+

Transient response
Temporary part
Decaying to zero

8.3 RLC电路的阶跃响应 (Step response of RLC circuits)

并联RLC电路



8.3 RLC电路的阶跃响应 (Step response of RLC circuits)

- Total step response of series RLC circuits

Overdamped: $v(t) = V_s + A_1 e^{s_1 t} + A_2 e^{s_2 t}$

Critically damped: $v(t) = V_s + (A_1 + A_2 t) e^{-\alpha t}$

Underdamped: $v(t) = V_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t}$

$$v(t) = v_{ss}(t) + v_t(t)$$



**Complete
response**

=

Steady-state response

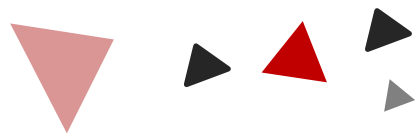
Permanent part $v(\infty)$
After all transients are done

+

Transient response

Temporary part
Decaying to zero

考前&考试注意事项



- 
1. 平时多练习，考前复盘一遍作业题（切勿轻敌）
 2. 考试时严格把控时间，切勿在一道题上花费过多时间
 3. 遇到不会的题目先做运算放大器
 4. 源变换YYDS,化简电路首选
 5. 过程>答案 如果没有时间，先写过程



← 非官方讨论群

